

# Tariff-Responsive Flexible Operation of a Full-Scale Reverse Osmosis Facility: Constraint-Aware Optimization and Annualized Economic Value

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## ABSTRACT

Electricity-price variability creates opportunities for reverse osmosis (RO) facilities to operate as flexible electrical loads, but the realizable economic value of this flexibility can be substantially affected by full-scale operational and tariff constraints. A deterministic mixed-integer linear programming framework was developed for the full-scale Orange County Water District RO facility to optimize hourly dispatch of heterogeneous parallel RO skids over a 168-h horizon while maintaining the required water production. The formulation accounted for product-water storage, peak-demand charges, startup and shutdown energy, skid availability, membrane fouling, and cleaning-related downtime. Six representative Southern California Edison time-of-use (TOU) and real-time-pricing (RTP) profiles were evaluated. Under the operationally constrained fixed-85% recovery case, weekly savings ranged from  $-1.0\%$  under conventional TOU pricing to  $62.3\%$  under the most volatile RTP profile. Progressively introducing full-scale constraints showed that

94.6 ML of product-water storage captured nearly all available within-week load-shifting value, whereas peak-demand charges produced the largest reduction from ideal flexible dispatch; startup/shutdown energy and the tested fouling and cleaning assumptions had smaller effects. Annualized fixed-85% savings ranged from 13.7% to 30.2% across the RTP scenarios, including USD 2.63 million yr<sup>-1</sup> (20.2%) under the base profile, compared with -0.8% under the TOU calendar. Allowing recovery to vary increased modeled savings but resulted in nearly exclusive selection of 90% recovery because per-skid power was held constant across recovery modes, indicating that this result is exploratory. Overall, the analysis distinguishes physical flexibility from economically realizable flexibility and identifies peak-demand exposure and the frequency of high-value electricity-price conditions as key determinants of full-scale RO flexibility value.

**Key Words** - reverse osmosis; flexible operation; mixed-integer linear programming; demand response; real-time electricity pricing; product-water storage; water-energy nexus

## 37 NOMENCLATURE

| Symbol                    | Definition                                                                                                                                                                            | Unit                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>SETS AND INDICES</b>   |                                                                                                                                                                                       |                                                                |
| $G$                       | Set of RO skid configurations, $G = \{\text{ERD, non-ERD}\}$ .                                                                                                                        | —                                                              |
| $t$                       | Hourly time index.                                                                                                                                                                    | —                                                              |
| $g$                       | RO skid-configuration index.                                                                                                                                                          | —                                                              |
| $r$                       | Discrete recovery-mode index.                                                                                                                                                         | —                                                              |
| $s$                       | Representative electricity-price/tariff index.                                                                                                                                        | —                                                              |
| $p$                       | Annual tariff-frequency-profile index.                                                                                                                                                | —                                                              |
| $o$                       | Operating-scenario index.                                                                                                                                                             | —                                                              |
| $\tau$                    | Auxiliary time index used in the active-cleaning summation.                                                                                                                           | —                                                              |
| <b>SYMBOLS</b>            |                                                                                                                                                                                       |                                                                |
| $A_{\text{clean}}$        | Number of skids actively undergoing cleaning.                                                                                                                                         | skid                                                           |
| $a_{f,b}$                 | Quadratic pump-curve coefficient for the feed pump (f) or interstage booster pump (b).                                                                                                | m (flow unit) <sup>-2</sup>                                    |
| $B_{\text{net}}$          | Annual net economic benefit in the investment-screening expression.                                                                                                                   | USD yr <sup>-1</sup>                                           |
| $C$                       | Cost; subscripts/superscripts identify total, energy, transition, cleaning, demand-charge, baseline, optimized, annual, weekly, storage, controls, cycling, or other cost components. | USD or USD yr <sup>-1</sup>                                    |
| $CF(R)$                   | RO concentration factor as a function of recovery ratio.                                                                                                                              | —                                                              |
| $c_t$                     | Electricity energy price at time t.                                                                                                                                                   | USD kWh <sup>-1</sup>                                          |
| $D$                       | Duration of one membrane-cleaning event.                                                                                                                                              | h                                                              |
| $D_t$                     | Downstream product-water demand at time t.                                                                                                                                            | ML d <sup>-1</sup>                                             |
| $E$                       | Event-related electrical energy; modifiers identify startup, shutdown, or cleaning.                                                                                                   | kWh event <sup>-1</sup>                                        |
| $F_{g,t}$                 | Aggregate fouling inventory for skid configuration g at time t.                                                                                                                       | equivalent skid-h                                              |
| $\Delta F_{\text{clean}}$ | Fouling-inventory reduction credited to one cleaning event.                                                                                                                           | equivalent skid-h                                              |
| $H$                       | Pump head; feed- and booster-pump variants are identified by subscripts f and b.                                                                                                      | m                                                              |
| $H_{0,f/b}$               | Assumed shutoff head for the feed or interstage booster pump.                                                                                                                         | m                                                              |
| $n_{g,t}$                 | Number of online skids in configuration g at time t; an additional recovery index r denotes recovery-specific dispatch.                                                               | skid                                                           |
| $N_g^{\text{max}}$        | Number of physically available skids in configuration g.                                                                                                                              | skid                                                           |
| $N_{\text{min/max}}$      | Minimum/maximum allowable total number of online duty skids.                                                                                                                          | skid                                                           |
| $P$                       | Electrical power; subscripts identify feed pump, booster pump, skid configuration, total RO demand, or peak demand.                                                                   | kW                                                             |
| $Q$                       | Volumetric flow or water-production rate; subscripts/superscripts identify feed, permeate, concentrate, fixed, rated, baseline, or recovery-specific quantities.                      | L h <sup>-1</sup> or ML d <sup>-1</sup>                        |
| $Q_{\text{target}}$       | Required total permeate production over the weekly optimization horizon.                                                                                                              | ML                                                             |
| $R$                       | RO recovery ratio; subscripts identify first-stage, reference, or discrete recovery-mode values.                                                                                      | —                                                              |
| $r_{\text{month/week}}$   | Monthly demand-charge rate or its prorated weekly equivalent.                                                                                                                         | USD kW <sup>-1</sup> month <sup>-1</sup> or week <sup>-1</sup> |
| $S_t$                     | Product-water storage inventory at time t.                                                                                                                                            | ML                                                             |
| $S_{\text{max}}$          | Maximum usable product-water storage capacity.                                                                                                                                        | ML                                                             |

| Symbol              | Definition                                                                                     | Unit            |
|---------------------|------------------------------------------------------------------------------------------------|-----------------|
| $SEC_g^{ref}$       | Reference specific energy consumption for skid configuration g.                                | $kWh\ m^{-3}$   |
| $T_{foul}$          | Reference fouling interval used to parameterize cleaning credit.                               | h               |
| $u_{g,t}$           | Number of skids started in configuration g at time t.                                          | skid            |
| $v_{g,t}$           | Number of skids shut down in configuration g at time t.                                        | skid            |
| $w_{p,s}$           | Equivalent number of weeks per year assigned to tariff s in annual profile p.                  | $week\ yr^{-1}$ |
| $z_{g,t}^{clean}$   | Number of skids in configuration g beginning a cleaning event at time t.                       | skid            |
| $\alpha_R$          | Recovery-dependent fouling multiplier in the fixed-recovery formulation.                       | —               |
| $\beta_g$           | Power-calibration factor for skid configuration g.                                             | —               |
| $\gamma$            | Exponent controlling the sensitivity of fouling accumulation to recovery/concentration factor. | —               |
| $\eta$              | Pump efficiency.                                                                               | —               |
| $\eta_{clean}$      | Fraction of accumulated fouling restored by one cleaning event.                                | —               |
| $\tau_{start/stop}$ | Assumed startup or shutdown duration.                                                          | s               |
| $\phi_r$            | Normalized fouling multiplier associated with recovery mode r.                                 | —               |

#### ABBREVIATIONS

|        |                                         |   |
|--------|-----------------------------------------|---|
| ERD    | Energy-recovery device.                 | — |
| GWRs   | Groundwater Replenishment System.       | — |
| MILP   | Mixed-integer linear programming.       | — |
| O&M    | Operation and maintenance.              | — |
| OCWD   | Orange County Water District.           | — |
| RO     | Reverse osmosis.                        | — |
| RTP    | Real-time pricing.                      | — |
| SCE    | Southern California Edison.             | — |
| SEC    | Specific energy consumption.            | — |
| TOU    | Time-of-use.                            | — |
| UV/AOP | Ultraviolet/advanced oxidation process. | — |

## 39 INTRODUCTION

40 Growing water scarcity, population growth, climate variability, and declining reliability of  
41 conventional freshwater supplies have increased the importance of reverse osmosis (RO) in  
42 desalination, potable water reuse, and advanced water purification (Warsinger et al., 2018). RO  
43 provides an effective barrier for dissolved salts (Wang et al., 2021), trace organic contaminants  
44 (Alhussaini et al., 2024), and other constituents that are difficult to remove through conventional  
45 treatment (Tang et al., 2018). This treatment capability has supported the development of  
46 increasingly large RO facilities (Eke et al., 2020); however, this benefit is accompanied by  
47 substantial electricity consumption (Kim et al., 2019). At full scale, the high-pressure pumps  
48 required to overcome osmotic pressure and membrane resistance typically make the RO process  
49 one of the largest electricity consumers within an advanced water-treatment facility (Voutchkov,  
50 2018). Consequently, electricity cost remains a major contributor to the overall cost of RO-based  
51 water production (Pearson et al., 2021).

52 The economic implications of RO electricity consumption depend not only on how much  
53 electricity is used, but also on when it is consumed (Zohrabian & Sanders, 2021). The growing  
54 contribution of variable renewable-energy resources has increased variability in net electricity  
55 demand and altered the temporal patterns of wholesale electricity prices, while grid-emission  
56 intensity also varies substantially across hours (De Chalendar et al., 2019; Mills et al., 2020).  
57 Utility rate structures increasingly reflect these dynamics through time-of-use pricing, real-time or  
58 other dynamic pricing mechanisms, and peak-demand charges (Chapin et al., 2024; Sowby, 2021).  
59 Consequently, the effective price of electricity can vary considerably by hour, day type, and season,  
60 with occasional high-price events substantially exceeding prices during surrounding periods  
61 (Chapin et al., 2024; Rao et al., 2024a). These dynamics create an opportunity for electricity-

intensive water-treatment processes to provide demand-side flexibility by shifting electricity consumption away from high-cost periods while continuing to satisfy water-production requirements (Rao, Atia, et al., 2024b; Rao, Bolorinos, et al., 2024).

Full-scale RO operation, however, cannot be shifted solely in response to electricity price. RO facilities are generally operated near stable production conditions to maintain membrane hydraulics, permeate production and quality, and equipment reliability (Mito et al., 2022; Ruiz-García et al., 2023). Although these priorities are operationally justified, they can produce electrical power demand profiles that are poorly aligned with time-varying electricity prices (Rao et al., 2024a). A facility operating at a nearly constant production rate may therefore consume similar amounts of electricity during both low- and high- price periods, even when downstream product-water storage or reserve treatment capacity could support temporal production shifting. As the electricity price variability increases, conventional steady-state production strategies may leave a substantial portion of the facility's potential economic flexibility unused (Ghorbani Bam et al., 2026; Rao, Atia, et al., 2024a).

Flexible RO operation involves more than simply reducing production during expensive hours and increasing it when electricity becomes cheaper. Full-scale facilities contain discrete treatment trains and parallel RO skids that must be started, stopped, and dispatched within equipment-availability and process-stabilization constraints. Individual units may also differ in energy efficiency because of differences in membrane and process configuration, interstage pumping, pump efficiency, and the presence or absence of energy-recovery devices (Park et al., 2020). Production shifting is constrained by downstream water demand and available product-water storage, while concentrating operation into fewer hours can increase peak-power demand and the resulting demand-charge component of the electricity bill (Chapin et al., 2024; Rao, Atia, et al.,

2024a). Frequent cycling introduces startup, production, shutdown, and flushing transitions, all of which may increase operational complexity and maintenance requirements (Jarma et al., 2024; Zhou et al., 2024). Membrane fouling and cleaning requirements can further restrict skid availability because fouling increases pressure and energy demand, whereas cleaning causes downtime and can reduce membrane service life (Hoek et al., 2022). Consequently, the theoretical value of electricity-price arbitrage may differ substantially from the savings achievable when realistic plant, tariff, and membrane-performance constraints are considered (Chapin et al., 2024).

Previous modeling and optimization studies have investigated flexible desalination and water-system operation through variable production, strategic water storage, renewable-energy integration, demand-response participation, and optimized pumping schedules (Elsir et al., 2023; Liu et al., 2022; Loutatidou et al., 2017; Mehrjerdi, 2020; Mkireb et al., 2019). However, these studies have generally addressed different subsets of the operational problem. Renewable-powered desalination studies have often allowed aggregate plant production to vary with available electricity while using water storage to decouple production from demand (Liu et al., 2022). Power–water coordination studies have similarly represented desalination facilities as aggregate electrical loads within day-ahead power-system scheduling models (Elsir et al., 2023; Mehrjerdi, 2020). Although these formulations provide important evidence of flexibility potential, they typically do not distinguish among parallel RO skids with different energy configurations or simultaneously represent finite product-water storage, demand charges, transition-energy requirements, fixed cycling penalties, fouling accumulation, and cleaning-related downtime. Moreover, short-term or day-ahead optimization results may not directly represent annual economic performance because electricity-price conditions, renewable availability, and high-cost events vary throughout the year (Rupiper et al., 2024). Consequently, an important unresolved

question is not simply whether an RO facility can operate flexibly, but how much of the theoretical flexibility value remains once full-scale physical and economic constraints are imposed, and which constraints control that value.

The full-scale Orange County Water District (OCWD) RO facility provides a suitable case study for addressing this question because its parallel RO architecture includes independently dispatchable units with different energy configurations, reserve treatment capacity, and downstream product-water storage. A deterministic mixed-integer linear programming (MILP) framework was therefore developed to optimize hourly dispatch of energy-recovery-device (ERD) and non-ERD RO skids while maintaining the required water production. Representative Southern California electricity price structures ranging from conventional time-of-use (TOU) schedules to more volatile real-time-pricing (RTP) were evaluated. Rather than considering only the optimized cost, the analysis progressively introduces individual physical and economic constraints to quantify how each constraint changes the economic value available under ideal flexible operation. Sensitivity analyses then evaluate the roles of storage, demand charges, cycling-related costs, fouling and cleaning assumptions, while an exploratory formulation examines discrete recovery-ratio flexibility. Finally, representative tariff-frequency scenarios are used to translate weekly optimization outcomes into annualized economic values.

Based on the physical and economic mechanisms described above, three hypotheses were evaluated:

- Tariff dependence: Flexible RO operation will provide little or no economic advantage when temporal electricity-price differences are small, whereas sufficiently volatile price signals will create substantially greater value for production shifting.

- Constraint hierarchy: The economic value predicted under ideal flexible dispatch will be reduced unevenly by full-scale constraints. Storage availability and peak-demand exposure are expected to exert stronger effects than short startup/shutdown transition-energy requirements and moderate fouling or cleaning constraints.
- Recovery and temporal scale: Recovery ratio flexibility is expected to provide additional operational value by changing permeate production per operating skid, but this benefit should be moderated by recovery-dependent membrane impacts. Moreover, annual economic value is expected to depend strongly on the frequency of favorable electricity-price conditions and therefore should not be inferred directly from the highest-saving individual weeks.

Testing these hypotheses allows the analysis to distinguish between operational flexibility and economically valuable flexibility at full scale. The contribution of this work is therefore not limited to estimating a potential percentage cost saving; rather, it identifies where the theoretical value of flexible RO operation is preserved or lost as realistic facility and tariff constraints are introduced, and examines how short-term flexibility translates into plant-level annual economic significance.

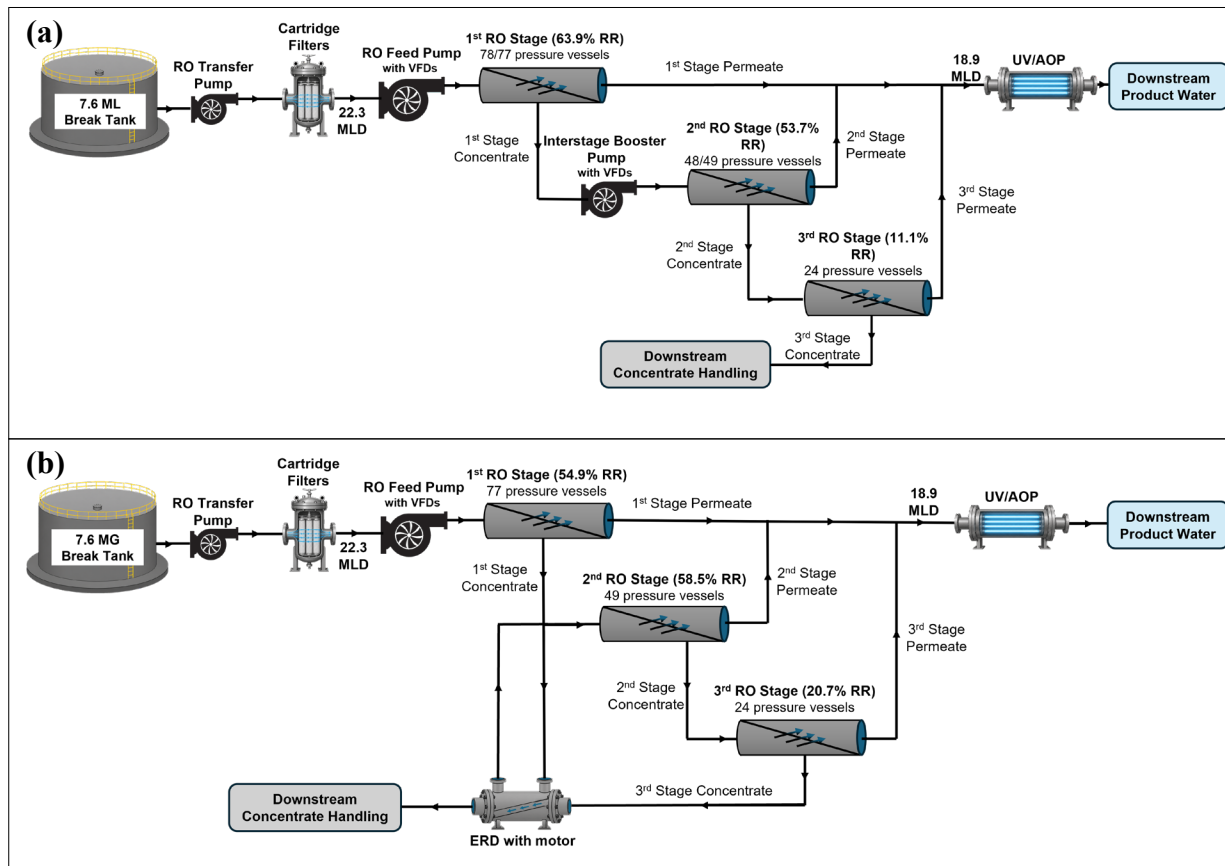
## **METHODS**

The methodology is organized in four main parts. First, the full-scale OCWD RO process configuration and the representative electricity-price structures used in the analysis are described. Second, the flexible-operation optimization framework is formulated, including skid dispatch, water-production and storage constraints, power demand, startup and shutdown transitions, demand charges, membrane fouling, and cleaning-related downtime. Third, a sequence of progressively constrained scenarios is defined to isolate the effects of individual operational and economic limitations on flexibility value. Finally, sensitivity analyses and annualization scenarios are introduced to evaluate the effects of key uncertain parameters and translate representative weekly optimization results into annual economic outcomes.

### **Facility and Process Details**

The Orange County Water District (OCWD) full-scale RO system consists of 27 parallel RO skids arranged within nine trains, designated A through I, with three skids assigned to each train. Of the 27 installed skids, 26 are available for duty operation and one is maintained as a standby unit. Each skid has a nominal permeate-production capacity of 18.9 million liters per day ( $\text{ML d}^{-1}$ ; 5 MGD); therefore, operation of all 26 duty skids provides a total nominal capacity of 492  $\text{ML d}^{-1}$  (130 MGD). The RO system is operated at approximately 85% recovery. Each skid is supplied by a dedicated RO feed pump with a design flow rate of 16,430  $\text{L min}^{-1}$  (4,340 gpm), a rated head of 207.6 m (681 ft), and a rated motor power of 745.7 kW (1,000 hp). Pretreated water is conveyed from the upstream break tank through transfer pumping and cartridge filtration before entering the high-pressure RO process. Permeate produced by the three RO stages is combined and subsequently conveyed to downstream UV/AOP treatment and product-water handling, while the final-stage concentrate is directed to the concentrate-management system, as illustrated in Fig. 1.

The installed RO skids include two principal hydraulic and energy configurations. Trains A-E contain 15 skids configured in a 78:48:24 pressure-vessel array without energy-recovery devices, whereas Trains F-G contain six skids configured in a 77:49:24 array and equipped with turbocharger energy-recovery devices. Trains H-I contain the remaining six skids, which are also configured in a 77:49:24 array but do not include energy-recovery devices. Interstage booster pumps are installed on the 21 non-ERD skids in Trains A-E and H-I to provide the additional pressure required by the downstream RO stages. Figure 1a presents the representative non-ERD configuration, in which first-stage concentrate is pressurized by an interstage booster pump before entering the second stage. Figure 1b presents the ERD-equipped configuration, in which hydraulic energy from the concentrate stream is recovered through a turbocharger. Based on the operational information provided by OCWD, the ERD-equipped configuration requires approximately 21.5 kW less power than the comparable non-ERD configuration (Veerapaneni et al., 2015), while the average specific energy consumption of the full-scale RO process is approximately  $0.54 \text{ kWh m}^{-3}$  (Gu et al., 2025). These two skid configurations were modeled separately because their different energy requirements affect the cost-optimal order in which the skids are dispatched.



**Figure 1.** Representative process configurations of the full-scale three-stage OCWD RO system: (a) non-ERD configuration with an interstage booster pump and (b) ERD-equipped configuration using a turbocharger to recover hydraulic energy from the concentrate stream. Permeate from all three stages is combined and conveyed to downstream UV/AOP treatment and product-water handling. The 7.6-ML upstream break tank shown in the schematic is distinct from the 94.6-ML downstream product-water storage represented in the flexible-operation model (RR = recovery ratio, MLD = million liter per day).

## **Tariff Structures**

OCWD currently procures electricity through a third-party contractual arrangement and is not directly billed under a Southern California Edison (SCE) retail tariff. Accordingly, the tariff structures evaluated in this study should not be interpreted as a reproduction of OCWD's current electricity contract but a potential alternate. Instead, representative SCE time-of-use and real-time-pricing structures were used as regional electricity price signals to assess how the economic value of flexible RO operation changes under different degrees of temporal price variability. This scenario-based approach allows the operational flexibility of the facility to be evaluated under price signals ranging from relatively predictable time-of-use schedules to highly volatile hourly pricing conditions.

As summarized in Table 1, six representative weekly electricity-price profiles were evaluated. The conventional TOU-8 profiles included separate summer and winter cases characterized by predefined on-peak and off-peak periods, together with peak-demand charges. The summer case contained a larger difference between on-peak and off-peak energy prices, whereas the winter case represented a less variable seasonal price structure. Four TOU-8-RTP profiles were used to represent hourly price signals with different combinations of seasonal conditions and weekend price severity. These included a Hot Summer Weekday + High-Cost Weekend profile, a Mild Summer Weekday + Low-Cost Weekend profile, a Winter Weekday + High-Cost Weekend profile, and a Winter Weekday + Low-Cost Weekend profile. The RTP HS+HW profile represented the most aggressive and volatile electricity-price condition, whereas the RTP W+LW profile represented comparatively mild hourly price variability. Together, these profiles provided a structured basis for evaluating the sensitivity of flexible RO operation to both seasonal tariff patterns and the magnitude of hourly electricity-price fluctuations.

218

219

**Table 1.** Representative Southern California Edison time-of-use and real-time-pricing structures used to

220

construct regional weekly electricity price signals for the flexible-operation analysis.

| SCE rate family | Representative weekly price profile            | Scenario Description                                                                                                                                                                                                                                                                                      |
|-----------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TOU-8           | Summer                                         | Representative conventional summer time-of-use profile with predefined on-peak, mid-peak, and off-peak energy-price periods and a peak-demand charge. The relatively large summer on-peak/off-peak price differential provides a moderate incentive for shifting RO production away from high-cost hours. |
|                 | Winter                                         | Representative conventional winter time-of-use profile with the same general rate architecture as the summer case but a smaller difference between high- and low-price periods. This profile represents lower seasonal grid stress and reduced energy-arbitrage potential.                                |
| TOU-8-RTP       | Hot Summer Weekday + High-Cost Weekend (HS+HW) | High-volatility summer RTP profile characterized by pronounced hourly price spikes during high-stress summer weekdays and elevated weekend prices. This profile represents the most aggressive price signal evaluated and provides an upper-end case for tariff-responsive operation.                     |
|                 | Mild Summer Weekday + Low-Cost Weekend (S+LW)  | Moderate-volatility summer RTP profile representing more typical summer weekday conditions and comparatively low weekend electricity prices. Hourly price differences remain sufficient to provide some opportunity for production shifting but are substantially smaller than under the HS+HW profile.   |
|                 | Winter Weekday + High-Cost Weekend (W+HW)      | Winter RTP profile representing generally lower seasonal electricity demand combined with an unusually high-cost weekend. This profile captures occasional winter price excursions and an intermediate level of electricity-price volatility.                                                             |
|                 | Winter Weekday + Low-Cost Weekend (W+LW)       | Comparatively low-volatility winter RTP profile with moderate weekday prices and a low-cost weekend. This profile represents relatively stable winter grid conditions and provides the smallest energy-arbitrage opportunity among the RTP cases.                                                         |

221

## 222 Flexible Operation Optimization

### 223 *Flexible RO operation problem formulation*

224 The flexible-operation problem was formulated as a deterministic mixed-integer linear  
225 programming (MILP) model over a one-week optimization horizon with hourly resolution. The  
226 horizon was defined as:

$$227 \quad t \in \mathcal{T} = \{1, 2, \dots, 168\}$$

228 with  $\Delta t = 1$  h. The objective of the optimization was to minimize the total weekly RO operating  
229 cost while maintaining the required weekly water production. The model considered the two  
230 physical RO skid configurations available at OCWD:

$$231 \quad g \in \mathcal{G} = \{\text{ERD}, \text{nonERD}\}$$

232 where ERD skids represent RO units equipped with turbocharger energy-recovery devices, while  
233 non-ERD skids represent conventional RO units with interstage booster pumps. The current  
234 OCWD configuration includes 27 total RO skids, of which 26 can be operated as duty skids, with  
235 6 ERD skids and 21 non-ERD skids available. The baseline operation was defined as 20  
236 continuously operating skids, consisting of 6 ERD skids and 14 non-ERD skids, producing  
237  $378.5 \text{ ML d}^{-1}$  at 85% recovery.

238 In this scenario, each online skid was assumed to operate at a fixed feed flow:

$$239 \quad Q_f^{\text{fix}} = \frac{Q_p^{\text{base}}}{R} \quad (1)$$

240 where:

$$241 \quad Q_p^{\text{base}} = 18.9 \text{ ML d}^{-1}$$

242  $R = 0.85$

243 Thus,

244 
$$Q_f^{\text{fix}} = \frac{18.9}{0.85} = 22.3 \text{ ML d}^{-1}$$

245 and the fixed permeate production from each online skid was:

246 
$$Q_p^{\text{fix}} = RQ_f^{\text{fix}} \tag{2}$$

247 Hence,  $Q_p^{\text{fix}} = 18.9 \text{ ML d}^{-1}$ . This formulation removes unrealistic continuous manipulation of  
248 skid-level flow and limits flexible operation to practical dispatch decisions: how many skids to  
249 operate, whether to prioritize ERD or non-ERD units, when to shift production through storage,  
250 and when skids must be shut down for cleaning.

251 *Decision variables*

252 The primary decision variable was the number of online skids of each configuration at each hour:

253 
$$n_{g,t} \in \mathbb{Z}_{\geq 0}$$

254 where  $n_{g,t}$  is the number of online skids in configuration  $g$  at hour  $t$ .

255 Additional integer variables were introduced to track startup, shutdown, and cleaning events:

256 
$$u_{g,t} \in \mathbb{Z}_{\geq 0}$$

257 
$$v_{g,t} \in \mathbb{Z}_{\geq 0}$$

258 
$$z_{g,t}^{\text{clean}} \in \mathbb{Z}_{\geq 0}$$

259 where  $u_{g,t}$  and  $v_{g,t}$  represent the number of skids started up and shut down, respectively, and  $z_{g,t}^{\text{clean}}$

260 represents the number of skids beginning a cleaning event at hour  $t$ .

261 Two continuous state variables were also included:

$$262 \quad S_t \geq 0$$

$$263 \quad F_{g,t} \geq 0$$

264 where  $S_t$  is product-water storage inventory and  $F_{g,t}$  is the aggregate fouling inventory for each  
265 skid configuration.

266 A continuous peak-power variable was introduced to represent the maximum power demand over  
267 the optimization horizon:

$$268 \quad p^{\text{peak}} \geq 0$$

269 The model was solved using *scipy.optimize.milp*, which is a SciPy interface to the HiGHS mixed-  
270 integer linear optimization solver.

271 *Fixed skid production and weekly water-production constraint*

272 Because per-skid production was fixed in this scenario, hourly RO permeate production was  
273 calculated directly from the number of online skids:

$$274 \quad Q_t^{\text{RO}} = \sum_{g \in \mathcal{G}} n_{g,t} Q_p^{\text{fix}} \quad (3)$$

275 The weekly production target was constrained to equal the baseline weekly production of 2,650  
276 ML:

$$277 \quad \sum_{t \in \mathcal{T}} Q_t^{\text{RO}} \Delta t = Q^{\text{target}} \quad (4)$$

278 where:

$$279 \quad Q^{\text{target}} = 378.5 \text{ ML d}^{-1} \cdot 7 \text{ d}$$

This constraint ensured that flexible operation changed the timing and configuration of production, but not the total weekly water volume delivered. Each weekly optimization was initialized from the steady-state baseline but was not required to return to the same online-skid configuration. Consequently, interweek transition costs were not included in the annualized calculation.

#### *Skid availability and online-skid constraints*

The total number of online skids was constrained between the minimum allowable online skid count and the maximum duty skid count:

$$N^{\min} \leq \sum_{g \in \mathcal{G}} n_{g,t} \leq N^{\max} \quad (5)$$

where:

$$N^{\min} = 3$$

$$N^{\max} = 26$$

Configuration-specific availability constraints were applied as:

$$0 \leq n_{\text{ERD},t} \leq 6$$

$$0 \leq n_{\text{nonERD},t} \leq 21$$

These constraints reflect the physical availability of ERD and non-ERD skids in the OCWD RO system.

#### *Pump-energy estimation*

Hourly power demand was calculated using fixed per-skid power coefficients for ERD and non-ERD configurations. For each online skid, the feed flow was fixed at:

$$Q_f^{\text{fix}} = 22.3 \text{ ML d}^{-1}$$

300 The feed-pump head was estimated using a quadratic pump-curve surrogate:

301 
$$H_f(Q_f) = H_{0,f} - a_f Q_f^2 \quad (6)$$

302 where  $H_{0,f}$  is the assumed feed-pump shutoff head and  $a_f$  is determined from the rated operating  
303 point. The feed-pump rated point was:

304 
$$Q_{f,r} = 16,430 \text{ L min}^{-1}$$

305 
$$H_{f,r} = 207.6 \text{ m}$$

306 and the shutoff head was assumed to be:

307 
$$H_{0,f} = 238.7 \text{ m}$$

308 Therefore:

309 
$$a_f = \frac{H_{0,f} - H_{f,r}}{Q_{f,r}^2} \quad (7)$$

310 For non-ERD skids, the interstage booster-pump flow was estimated from the first-stage  
311 concentrate flow:

312 
$$Q_b = (1 - R_1) Q_f^{\text{fix}} \quad (8)$$

313 where:

314 
$$R_1 = 0.639$$

315 The booster-pump head was similarly represented as:

316 
$$H_b(Q_b) = H_{0,b} - a_b Q_b^2 \quad (9)$$

317 where  $H_{0,b} = 28.8 \text{ m}$ , and  $a_b$  was calculated from the nominal booster-pump operating point.

318 Pump power was calculated from flow and head using:

$$319 \quad P = \frac{QHSG}{3600\eta} \quad (10)$$

320 where P is pump power in kW, Q is flow in L h<sup>-1</sup>, H is head in m, SG is specific gravity, and  $\eta$  is  
321 pump efficiency.

322 For ERD skids, the raw hydraulic power was calculated using the feed-pump surrogate and then  
323 calibrated to the reference ERD SEC. For non-ERD skids, total raw hydraulic power included both  
324 feed-pump and interstage-booster power:

$$325 \quad P_{\text{nonERD}}^{\text{raw}} = P_f + P_b \quad (11)$$

326 The raw pump-power values were scaled so that the model matched the reference SEC values at  
327 the 85% recovery and 18.9 ML d<sup>-1</sup>/skid operating point:

$$328 \quad SEC_{\text{ERD}} = 0.519 \text{ kWh m}^{-3}$$

$$329 \quad SEC_{\text{nonERD}} = 0.546 \text{ kWh m}^{-3}$$

330 The calibration factor for each group was:

$$331 \quad \beta_g = \frac{SEC_g^{\text{ref}} Q_p^{\text{fix}}}{P_g^{\text{raw}}} \quad (12)$$

332 and the calibrated per-skid power was:

$$333 \quad P_g = \beta_g P_g^{\text{raw}} \quad (13)$$

334 The hourly RO power demand was then:

$$P_t = \sum_{g \in \mathcal{G}} P_g n_{g,t} \quad (14)$$

The calibrated power coefficients were configuration-specific but not recovery-specific. Accordingly, the same per-skid power coefficient was applied to a given ERD or non-ERD skid at 80%, 85%, and 90% recovery. Thus, in the recovery-flexible formulation, changing recovery affected permeate production and fouling accumulation but did not directly change the modeled instantaneous power demand of an online skid.

#### *Startup and shutdown constraints*

To discourage unrealistic hour-to-hour switching, startup and shutdown variables were introduced at the physical configuration level. The transition balance was:

$$n_{g,t} - n_{g,t-1} = u_{g,t} - v_{g,t} \quad (15)$$

For the first hour, the transition equation was initialized using the baseline skid counts:

$$N_{\text{ERD},0} = 6$$

$$N_{\text{nonERD},0} = 14$$

Startup and shutdown energy penalties were calculated from the nominal per-skid power and assumed startup/shutdown durations:

$$E_g^{\text{start}} = P_g \frac{\tau^{\text{start}}}{3600}$$

$$E_g^{\text{stop}} = P_g \frac{\tau^{\text{stop}}}{3600}$$

where the base-case values were:

$$\tau^{\text{start}} = 200 \text{ s}$$

$$\tau^{\text{stop}} = 60 \text{ s}$$

The corresponding transition cost was:

$$C^{\text{trans}} = \sum_{t \in T} \sum_{g \in G} c_t (E_g^{\text{start}} u_{g,t} + E_g^{\text{stop}} v_{g,t}) \quad (16)$$

where  $c_t$  is the hourly electricity price. In the base model, startup and shutdown events were represented only through the additional electricity consumed during the transition periods. The startup and shutdown energy requirements were calculated from the nominal per-skid power and assumed transition durations of 200 s and 60 s, respectively. No additional fixed monetary penalties were assigned to startup or shutdown events in the base case because site-specific information regarding equipment wear, maintenance requirements, labor, and loss of membrane life was unavailable. Consequently, the base-case transition-cost component reflects only transition-related energy cost. Fixed startup and shutdown penalty costs were introduced separately in the sensitivity analysis to evaluate how unmodeled cycling-related costs could affect the economic value of flexible operation.

For the startup/shutdown fixed-cost sensitivity analysis, the transition-cost formulation was extended to include user-specified monetary penalties per startup and shutdown event:

$$C_{\text{sens}}^{\text{trans}} = \sum_{t \in T} \sum_{g \in G} [c_t (E_g^{\text{start}} u_{g,t} + E_g^{\text{stop}} v_{g,t}) + C_g^{\text{start,fix}} u_{g,t} + C_g^{\text{stop,fix}} v_{g,t}] \quad (17)$$

where  $C_g^{\text{start,fix}}$  and  $C_g^{\text{stop,fix}}$  are the assumed fixed monetary penalties associated with starting and shutting down one skid, respectively. These parameters were set to zero in the base model and were assigned nonzero values only in the dedicated sensitivity analysis.

373 *Product-water storage balance*

374 Product-water storage was included to allow load shifting while maintaining downstream water  
375 demand. The storage balance was:

376 
$$S_t = S_{t-1} + (Q_t^{RO} - D_t)\Delta t \quad (18)$$

377 where  $D_t$  is downstream demand. In the absence of an external demand profile,  $D_t$  was assumed  
378 to be constant and equal to 378.5 ML d<sup>-1</sup>.

379 Storage inventory was constrained by:

380 
$$0 \leq S_t \leq S^{\max}$$

381 where the base-case storage capacity was equal to 6 hours of 378.5 ML d<sup>-1</sup> operation:

382 
$$S^{\max} = 94.6 \text{ ML}$$

383 The initial storage was set to 50% of capacity:

384 
$$S_0 = 0.5S^{\max}$$

385 A terminal storage constraint was imposed to prevent the optimizer from depleting storage at the  
386 end of the horizon:

387 
$$S_T = S_0$$

388 This ensured that load shifting was achieved through within-week redistribution of RO production  
389 rather than by ending the optimization horizon with an artificially depleted tank.

390 *Demand-charge formulation*

391 To represent peak-demand charges, the model introduced an auxiliary peak-power variable:

$$p^{\text{peak}} \geq P_t \forall t \in \mathcal{T} \quad (19)$$

The weekly demand-charge cost was:

$$C^{\text{demand}} = r^{\text{week}} p^{\text{peak}} \quad (20)$$

where  $r^{\text{week}}$  is the prorated weekly demand-charge rate:

$$r^{\text{week}} = r^{\text{month}} \frac{7}{30}$$

The base-case monthly demand-charge rate was set to:

$$r^{\text{month}} = 20 \text{ USD/kW-month}$$

This term penalized operating strategies that reduce energy cost by concentrating production into a small number of high-power hours.

Because the optimization horizon was one week, the monthly demand-charge rate was converted to an equivalent weekly rate. This approximation preserves the economic penalty associated with peak power within each representative week but does not reproduce the interaction among weekly peaks within a monthly billing period.

#### *Fouling and cleaning formulation*

An aggregate fouling inventory was tracked separately for ERD and non-ERD skids. RO membrane fouling is known to reduce permeability and increase operating pressure and cleaning frequency; therefore, a fouling-state constraint was included to prevent the optimizer from continuously favoring one skid group without accounting for cleaning requirements.

The fouling inventory for each configuration was expressed in equivalent full-load skid-hours at the reference recovery:  $F_{g,t}$

Because the current scenario operates at 85% recovery, the recovery-dependent fouling multiplier is equal to unity. More generally, the fouling multiplier was defined from the RO concentration factor:

$$CF(R) = \frac{1}{1-R} \quad (21)$$

$$\alpha_R = \left[ \frac{CF(R)}{CF(R_{ref})} \right]^\gamma \quad (22)$$

where  $R_{ref} = 0.85$  and  $\gamma$  is a tunable exponent. In this fixed-recovery scenario:

$$R = R_{ref} = 0.85$$

$$\alpha_R = 1$$

The power-law relationship between concentration factor and fouling accumulation is a heuristic scaling relationship rather than an empirically calibrated fouling law for the OCWD facility. The exponent therefore controls the assumed sensitivity of fouling to recovery and was varied systematically in the sensitivity analysis.

The fouling balance was:

$$F_{g,t} = F_{g,t-1} + \Delta t \alpha_R n_{g,t} - C^{clean} z_{g,t-D+1}^{clean} \quad (23)$$

where  $D$  is the cleaning duration in hours and  $C^{clean}$  is the fouling removal credit from one cleaning event. The cleaning duration was set to:

$$D = 4 \text{ h}$$

and the cleaning credit was:

$$F^{clean} = \eta^{clean} T^{foul} \quad (24)$$

431 where  $\eta^{\text{clean}}$  is the restoration fraction and  $T^{\text{foul}}$  is the reference fouling interval. The base-case  
 432 values were:

$$433 \quad \eta^{\text{clean}} = 1.0$$

$$434 \quad T^{\text{foul}} = 720 \text{ h}$$

435 Cleaning was modeled as a downtime event. If a skid was undergoing cleaning, it was unavailable  
 436 for production. The number of skids being cleaned at any given time (i.e., active cleaning counts)  
 437 was defined as:

$$438 \quad A_{g,t}^{\text{clean}} = \sum_{\tau=t-D+1}^t z_{g,\tau}^{\text{clean}} \quad (25)$$

439 and the availability constraint became:

$$440 \quad n_{g,t} + A_{g,t}^{\text{clean}} \leq N_g^{\text{max}} \quad (26)$$

441 where  $N_g^{\text{max}}$  is the number of available skids in group g. This formulation ensures that cleaning  
 442 occurs during offline periods and prevents the optimizer from assigning the same skid capacity to  
 443 production and cleaning simultaneously. Manufacturer guidance commonly recommends RO  
 444 cleaning based on normalized performance deterioration, including increased pressure drop or  
 445 decreased normalized permeate flow, supporting the inclusion of cleaning as an operational  
 446 constraint rather than a purely economic penalty.

447 The fouling inventory was bounded as:

$$448 \quad 0 \leq F_{g,t} \leq N_g^{\text{max}} T^{\text{foul}} \quad (27)$$

449 and a terminal fouling constraint was imposed:

$$F_{g,T} \leq F_{g,0} \quad (28)$$

This terminal condition prevents the optimizer from accumulating fouling near the end of the week and deferring the cleaning burden beyond the modeled horizon.

### *Objective function*

The MILP minimized total weekly operating cost  $\min C^{\text{total}}$ , with:

$$C^{\text{total}} = C^{\text{energy}} + C^{\text{trans}} + C^{\text{clean}} + C^{\text{demand}} \quad (29)$$

The energy-cost component was:

$$C^{\text{energy}} = \sum_{t \in \mathcal{T}} c_t P_t \Delta t \quad (30)$$

The startup/shutdown cost was:

$$C^{\text{trans}} = \sum_{t \in \mathcal{T}} \sum_{g \in \mathcal{G}} c_t (E_g^{\text{start}} u_{g,t} + E_g^{\text{stop}} v_{g,t}) \quad (31)$$

For the base-case formulation,  $C^{\text{trans}}$  included only the cost of additional electricity consumed during startup and shutdown transitions. Fixed monetary penalties associated with equipment cycling were set to zero and therefore did not contribute to the base-case objective function. Nonzero fixed startup and shutdown penalties were included only when conducting the cycling-cost sensitivity analysis.

The cleaning cost was:

$$C^{\text{clean}} = \sum_{t \in \mathcal{T}} \sum_{g \in \mathcal{G}} z_{g,t}^{\text{clean}} (C_g^{\text{clean, fixed}} + c_t E_g^{\text{clean}}) \quad (32)$$

The demand-charge component was:

$$C^{\text{demand}} = r^{\text{week}} p^{\text{peak}} \quad (33)$$

Thus, the base model identified the least-cost weekly operating schedule while satisfying water-production, skid-availability, storage, startup/shutdown, demand-charge, and fouling/cleaning constraints. Startup and shutdown events affected the base objective through their transition-related energy consumption cost, while additional fixed cycling penalties were excluded unless explicitly introduced in the sensitivity analysis.

#### *Baseline comparison*

The optimized schedules were compared against a non-flexible baseline in which 20 skids operated continuously at 85% recovery and 18.9 ML d<sup>-1</sup>/skid:

$$\begin{aligned} n_{\text{ERD},t}^{\text{base}} &= 6 \\ n_{\text{nonERD},t}^{\text{base}} &= 14 \\ Q_t^{\text{base}} &= 378.5 \text{ ML d}^{-1} \end{aligned}$$

The baseline monetary cost included energy and demand-charge components. Direct cleaning costs were set to zero in both the baseline and optimized cases. Fouling and cleaning were represented operationally through accumulated equivalent skid-hours, cleaning-related downtime, and skid-availability constraints rather than through a direct base-case cleaning expenditure. Savings were calculated as:

$$\text{Savings} = C^{\text{base}} - C^{\text{opt}} \quad (34)$$

$$\text{Savings}(\%) = 100 \frac{C^{\text{base}} - C^{\text{opt}}}{C^{\text{base}}} \quad (35)$$

*Scenario design for constraint-isolation and recovery-flexibility analysis*

A structured scenario analysis was performed to evaluate how individual physical and economic constraints affected the achievable value of flexible RO operation. All scenarios were evaluated over the same one-week optimization horizon with hourly resolution and were required to satisfy the same weekly permeate-production target of 2,650 ML. The non-flexible comparison baseline consisted of 20 continuously operating skids, six ERD and 14 non-ERD skids, producing 378.5 ML d<sup>-1</sup> at 85% recovery.

Scenario L0 was defined as the operationally constrained reference case. It included separate dispatch of ERD and non-ERD skids, 94.6 ML of usable product-water storage, demand charges, startup and shutdown energy penalties, fouling and cleaning constraints, and a fixed recovery ratio of 85%. Scenarios L1–L4 were then used as an auxiliary constraint-isolation sequence to determine how progressively introducing individual operational constraints changed the economic value of flexibility. L1 represented the ideal weekly dispatch case, subject only to skid availability and total weekly water production. L2 added the finite product-water storage balance, L3 added the peak-demand charge, and L4 added startup and shutdown transition-energy costs based on the assumed startup and shutdown durations. Additional fixed monetary penalties for cycling were not included in L4 or in the operationally constrained L0 base case. The effect of fouling and cleaning constraints was therefore isolated by comparing L4 with the fully constrained L0 case.

Scenario L5 extended the operationally constrained L0 formulation by allowing each operating skid to select among discrete recovery ratios of 80%, 85%, and 90%. Consequently, the numerical labels L0–L5 do not represent a strictly sequential progression from the least constrained to the most constrained formulation. Instead, L0 is the operationally constrained reference case, L1–L4

509 constitute the constraint-build sequence, and L5 evaluates recovery-ratio flexibility relative to L0.

510 The scenario definitions are summarized in Table 2.

511

**Table 2.** Definitions of the reference, constraint-isolation, and recovery-flexibility scenarios used in the weekly MILP analysis.

| Scenario | Description                                                                                                                                                                                                                                                                                       | Analytical purpose                                                                                                              |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| L0       | Operationally constrained fixed-85% reference case including ERD/non-ERD skid dispatch, 94.6 ML product-water storage, demand charges, startup/shutdown transition-energy costs, and fouling and cleaning constraints. Additional fixed monetary startup and shutdown penalties were set to zero. | Represents the complete and operationally constrained reference case formulation.                                               |
| L1       | Ideal weekly dispatch subject to ERD/non-ERD skid availability and the 2,650 ML weekly production requirement, without a finite storage limit, demand charges, startup/shutdown penalties, or fouling and cleaning constraints.                                                                   | Establishes the upper-bound economic value of flexible dispatch.                                                                |
| L2       | L1 plus a 94.6 ML product-water storage balance, initial storage condition, and terminal storage equality.                                                                                                                                                                                        | Isolates the effect of finite downstream storage on load-shifting potential.                                                    |
| L3       | L2 plus the peak-demand-charge component.                                                                                                                                                                                                                                                         | Quantifies the reduction in savings caused by peak-power exposure.                                                              |
| L4       | L3 plus startup and shutdown transition-energy costs based on assumed transition durations of 200 s and 60 s. Fixed monetary startup and shutdown penalties were set to zero.                                                                                                                     | Isolates the effect of operational cycling and transition energy requirements.                                                  |
| L5       | L0 plus discrete recovery-ratio flexibility at 80%, 85%, and 90%.                                                                                                                                                                                                                                 | Evaluates whether recovery flexibility provides additional value beyond the operationally constrained fixed-85% reference case. |

For the fixed-recovery scenarios, all online skids operated at the nominal recovery ratio of 85%. In the recovery-flexible scenario, each online skid was assigned one of three discrete recovery modes: 80%, 85%, or 90%. Feed flow was held constant at 22.3 ML d<sup>-1</sup> per online skid for all three modes, and permeate production was calculated as:

$$Q_{g,r,t}^P = R_r Q^F n_{g,r,t} \quad (36)$$

where  $R_r$  is the selected recovery ratio. The corresponding permeate-production rates were 17.8, 18.9, and 20.0 ML d<sup>-1</sup> per online skid at 80%, 85%, and 90% recovery, respectively. Concentrate flow therefore decreased as recovery increased according to  $Q^C = Q^F(1 - R_r)$ .

The calibrated per-skid power coefficients for ERD and non-ERD configurations were held constant across the three recovery modes. In other words, the model did not independently recalculate feed pressure, booster-pump power, or ERD performance as recovery changed. Consequently, higher recovery increased modeled water production per unit of assigned skid power. This simplifying assumption was necessary because recovery-specific hydraulic and energy-performance data were unavailable for the full-scale facility. Potential increases in osmotic pressure, membrane pressure requirement, salt passage, scaling risk, and other recovery-dependent hydraulic effects were not modeled explicitly.

Recovery-dependent membrane stress was represented through the fouling formulation. The concentration factor for recovery mode  $r$  was defined as:

$$CF_r = \frac{1}{1-R_r} \quad (37)$$

and the normalized fouling multiplier was calculated relative to the nominal 85% recovery condition:

$$\phi_r = \left( \frac{CF_r}{CF_{85}} \right)^\alpha \quad (38)$$

where  $\alpha$  is the fouling-recovery exponent and  $CF_{85} = 1/(1 - 0.85)$ . The reference case used  $\alpha = 1.5$ , resulting in fouling multipliers of approximately 0.65, 1.00, and 1.84 at 80%, 85%, and 90% recovery, respectively. Thus, under the adopted model, fouling accumulated approximately 2.83

times faster at 90% recovery than at 80% recovery. These values represent model-based relative fouling rates rather than experimentally measured full-scale fouling rates.

In addition to the constraint-isolation scenarios, sensitivity analyses were performed for product-water storage capacity, demand-charge rate, fouling interval, cleaning duration, fouling-recovery exponent, and fixed startup/shutdown costs. Storage capacity was varied from 0 to 189.3 ML; the demand-charge rate was varied from 0 to 60 USD/(kW. month); the fouling interval was varied from 360 to 1,440 h; cleaning duration was varied from 2 to 8 h; and the fouling-recovery exponent was varied from 1.0 to 2.0. Because fixed monetary cycling penalties were not included in the base model, an additional sensitivity analysis assigned startup/shutdown penalty pairs ranging from 0/0 to 250/50 USD per skid. These values were added to the transition-energy cost and were used to evaluate the potential influence of equipment wear, maintenance, labor, and other cycling-related impacts that could not be quantified from the available site data.

#### *Annualized value analysis under tariff-frequency assumptions*

Because the optimization horizon was one week, an annualized analysis was performed to translate weekly tariff-specific savings into yearly economic value. This step was included to avoid over-interpreting the most extreme RTP tariff case as if it occurred every week of the year. Annual cost was calculated by weighting each representative weekly tariff structure by an equivalent number of weeks per year:

$$C_{p,o}^{\text{annual}} = \sum_{s \in S} w_{p,s} C_{s,o}^{\text{week}} \quad (39)$$

where  $C_{p,o}^{\text{annual}}$  is the annual cost for annualization profile  $p$  and operating scenario  $o$ ,  $w_{p,s}$  is the equivalent number of weeks per year assigned to tariff structure  $s$ , and  $C_{s,o}^{\text{week}}$  is the optimized

weekly cost for tariff structure  $s$  under operating scenario  $o$ . The corresponding annual baseline cost was calculated as:

$$C_p^{\text{base,annual}} = \sum_{s \in S} w_{p,s} C_s^{\text{base,week}} \quad (40)$$

Annual savings were then calculated as:

$$S_{p,o}^{\text{annual}} = C_p^{\text{base,annual}} - C_{p,o}^{\text{annual}} \quad (41)$$

and the annual percentage savings were calculated as:

$$S_{p,o}^{\text{annual}}[\%] = 100 \frac{C_p^{\text{base,annual}} - C_{p,o}^{\text{annual}}}{C_p^{\text{base,annual}}} \quad (42)$$

where the annual weighting horizon was  $365/7 = 52.14$  equivalent weeks  $\text{yr}^{-1}$ .

Four annualization profiles were evaluated. The first profile, termed the TOU calendar, represented operation under the conventional TOU-8 calendar, with 17.43 equivalent summer weeks and 34.71 equivalent winter weeks. The other three profiles represented RTP-based price-signal cases with different levels of price volatility: RTP conservative, RTP base, and RTP high-volatility. In these profiles, the annual equivalent weeks were distributed among the representative RTP tariff structures: Hot Summer Weekday + High-Cost Weekend, Mild Summer Weekday + Low-Cost Weekend, Winter Weekday + High-Cost Weekend, and Winter Weekday + Low-Cost Weekend. The RTP conservative profile assigned fewer weeks to high-cost RTP structures, whereas the high-volatility profile assigned more weeks to high-cost RTP conditions (Table 3).

**Table 3.** Equivalent-week weights used for annualized tariff profiles.

| <b>Annual profile</b> | <b>TOU-S</b> | <b>TOU-W</b> | <b>RTP<br/>HS+HW</b> | <b>RTP<br/>MS+LW</b> | <b>RTP<br/>W+HW</b> | <b>RTP<br/>W+LW</b> |
|-----------------------|--------------|--------------|----------------------|----------------------|---------------------|---------------------|
| RTP base              | 0.00         | 0.00         | 4.00                 | 13.43                | 6.00                | 28.71               |
| RTP conservative      | 0.00         | 0.00         | 2.00                 | 15.43                | 4.00                | 30.71               |
| RTP high-volatility   | 0.00         | 0.00         | 8.00                 | 9.43                 | 10.00               | 24.71               |
| TOU calendar          | 17.43        | 34.71        | 0.00                 | 0.00                 | 0.00                | 0.00                |

Annualized values were calculated for two operating strategies: the operationally constrained fixed-85% reference case and the recovery-flexible 80/85/90% case. Both cases included 94.6 ML product-water storage, demand charges, startup/shutdown penalties, and fouling/cleaning constraints. Therefore, the annualized analysis did not introduce a new optimization formulation; instead, it aggregated the weekly MILP results using tariff-frequency weights. The RTP frequency profiles were constructed as representative scenarios rather than forecasts of OCWD's electricity contract or empirically observed annual SCE billing frequencies. Each profile sums to 52.14 equivalent weeks, while the allocation among the four RTP price signals was varied to examine the influence of increasingly frequent high-price conditions on annual economic value.

## RESULTS

The results are presented in four stages. First, the economic and operational responses of the fixed-85% reference case are compared across the six representative electricity-price structures. Second, the constraint-isolation and recovery-flexibility analyses are used to determine how individual physical and economic constraints alter the optimized operating cost. Third, sensitivity analyses examine the influence of product-water storage, demand charges, fouling and cleaning assumptions, and startup/shutdown cycling costs. Finally, the representative weekly results are aggregated using tariff-frequency scenarios to evaluate their annual economic significance. Where relevant, the findings are interpreted relative to the hypotheses introduced above to distinguish expected behavior from outcomes that provide additional insight into the limiting mechanisms of flexible RO operation.

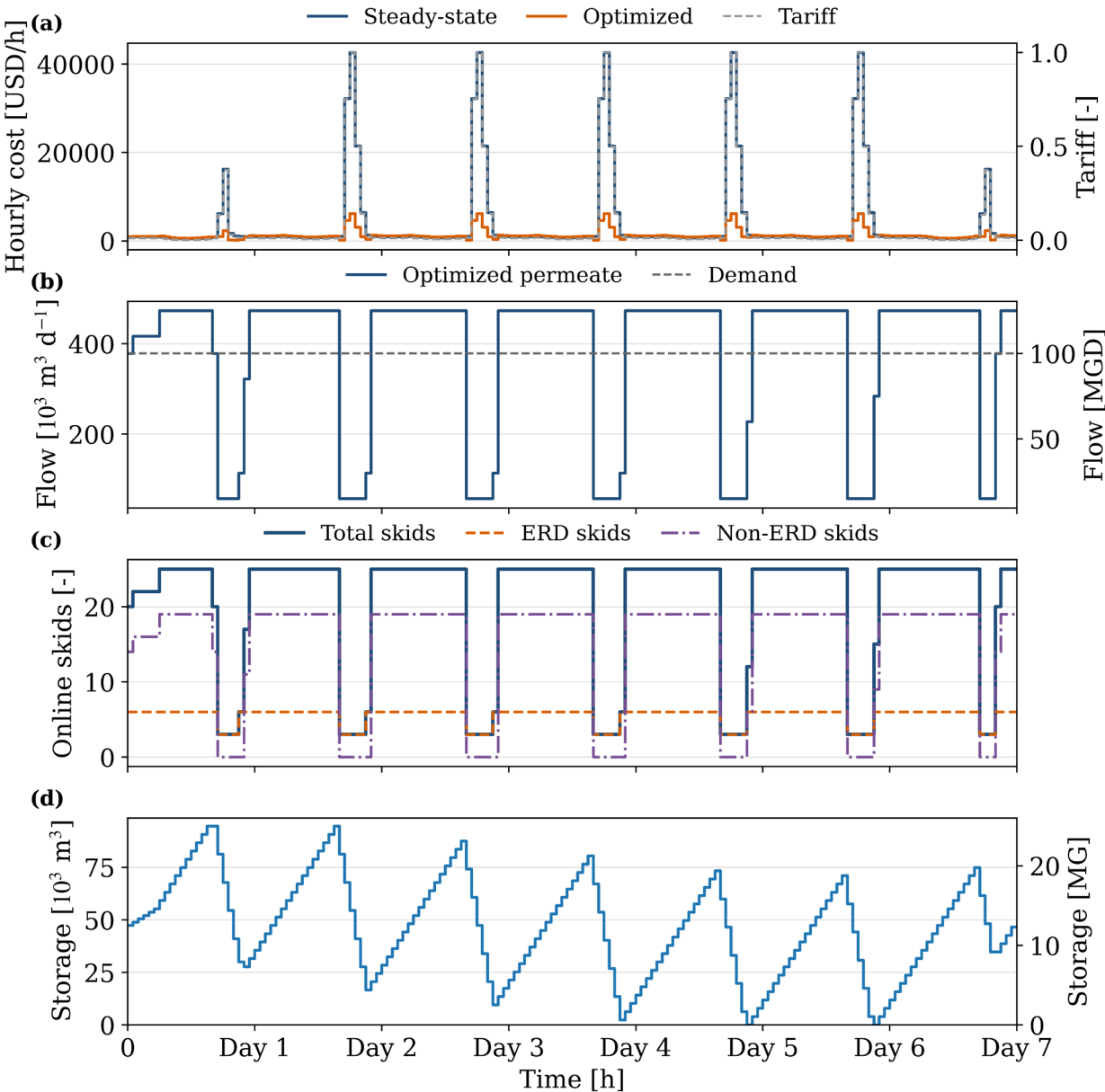
### **Tariff-Dependent Economic Value of Flexible RO Operation**

Flexible RO operation showed highly tariff-dependent economic value (Table 4). Under the operationally constrained fixed-85% reference case with storage, demand charges, startup/shutdown penalties, and fouling/cleaning constraints, savings ranged from slightly negative values under conventional TOU tariffs to large reductions under volatile RTP tariffs. Specifically, the fixed-85% case produced savings of -0.4% and -1.0% under TOU-8 summer and TOU-8 winter, respectively, indicating that conventional TOU price spreads were insufficient to overcome operational constraints. In contrast, savings reached 62.3% under the Hot Summer Weekday + High-Cost Weekend tariff, 27.3% under the Winter Weekday + High-Cost Weekend tariff, and 10.6% under the Mild Summer Weekday + Low-Cost Weekend tariff.

**Table 4.** Weekly cost savings under the fixed-85% recovery case.

| <b>Tariff</b> | <b>Baseline cost<br/>[USD]</b> | <b>Optimized cost<br/>[USD]</b> | <b>Savings<br/>[USD]</b> | <b>Savings<br/>[%]</b> | <b>Peak<br/>power<br/>[kW]</b> | <b>Total online<br/>skids (min–max)</b> |
|---------------|--------------------------------|---------------------------------|--------------------------|------------------------|--------------------------------|-----------------------------------------|
| TOU-8 Summer  | 209,884                        | 210,768                         | –883                     | –0.4                   | 9,776                          | 6-23                                    |
| TOU-8 Winter  | 198,807                        | 200,711                         | –1,903                   | –1.0                   | 8,915                          | 19-21                                   |
| RTP HS+HW     | 726,019                        | 273,628                         | 452,391                  | 62.3                   | 10,637                         | 3-25                                    |
| RTP MS+LW     | 220,840                        | 197,532                         | 23,308                   | 10.6                   | 9,776                          | 3-23                                    |
| RTP W+HW      | 280,395                        | 203,903                         | 76,492                   | 27.3                   | 9,776                          | 3-23                                    |
| RTP W+LW      | 191,044                        | 189,186                         | 1,858                    | 1.0                    | 9,345                          | 3-22                                    |

These results indicate that flexible operation is not universally cost-effective. Rather, its value depends on the magnitude and duration of electricity-price differentials. When price volatility was limited, the optimizer had little opportunity to shift production without increasing demand charges or triggering additional cycling. Conversely, under high-price RTP events, the optimizer substantially reduced production during high-cost hours and compensated during lower-cost periods using storage and available skid capacity.



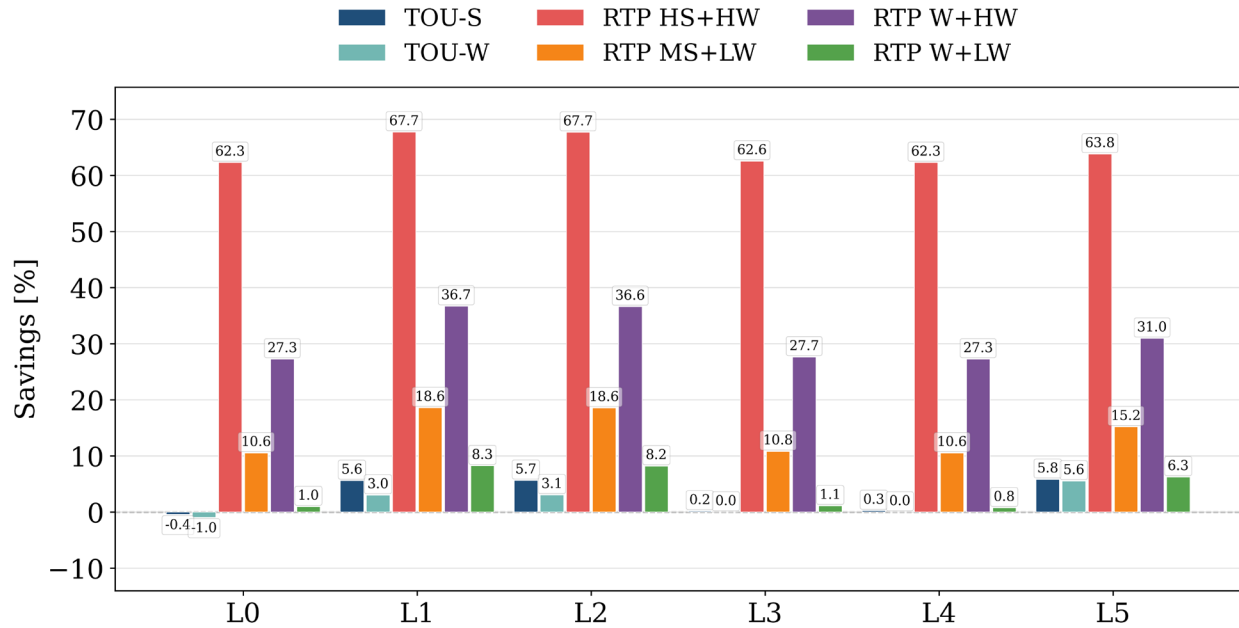
630 **Figure 2.** Optimized hourly operation under the RTP Hot Summer Weekday + High-Cost Weekend price  
631 signal for the operationally constrained fixed-85% reference case: (a) steady-state and optimized hourly  
632 electricity costs together with the normalized electricity price signal; (b) optimized permeate production  
633 and constant downstream demand; (c) total, ERD, and non-ERD online skid counts; and (d) downstream  
634 product-water storage inventory.

## **Effect of Operational Constraints and Recovery Flexibility on Achievable Savings**

Figure 3 compares the operationally constrained fixed-85% reference case, L0, with the auxiliary constraint-isolation sequence L1–L4 and the recovery-flexible case L5. L1 represents an ideal upper bound subject only to skid availability and the weekly production requirement. Under L1, savings reached 67.7% for RTP HS+HW, 36.7% for RTP W+HW, 18.6% for RTP MS+LW, and 8.3% for RTP W+LW, compared with 5.6% and 3.0% under TOU-8 Summer and Winter, respectively.

Introducing 94.6 ML of finite product-water storage in L2 produced results nearly identical to those of L1, indicating that this storage capacity captured most of the available within-week load-shifting potential. The largest reduction occurred between L2 and L3, when demand charges were introduced. RTP HS+HW savings decreased from 67.7% to 62.6%, while RTP W+HW savings decreased from 36.6% to 27.7%. Adding startup and shutdown transition-energy costs in L4 caused a smaller additional reduction.

The L4-to-L0 comparison isolates the effect of fouling and cleaning constraints, which had little influence on the high-volatility RTP cases but affected scenarios with limited underlying economic value. The L0-to-L5 comparison shows the modeled value of recovery flexibility under fixed-feed and fixed-power assumptions. Recovery flexibility increased savings across all six price signals, although the increase should be interpreted as exploratory.



**Figure 3.** Weekly cost savings across the reference, constraint-isolation, and recovery-flexibility scenarios for the six representative electricity price structures. Scenarios are displayed in numerical order from L0 to L5. L0 represents the fully constrained fixed-85% reference case; L1–L4 progressively introduce finite storage, demand charges, and startup and shutdown transition-energy costs from the ideal dispatch case; and L5 extends L0 with discrete recovery-ratio flexibility. The effect of fouling and cleaning constraints is isolated by comparing L4 with L0.

The constraint-isolation analysis identifies demand-charge exposure as the largest economic limitation on flexible RO operation. The substantial reduction between L2 and L3 shows that concentrating production during low-energy-price periods can increase peak-power costs and thereby reduce the value of energy arbitrage. The smaller difference between L3 and L4 indicates that startup and shutdown energy penalties had a more limited effect under the base-case assumptions. Similarly, the L4-to-L0 comparison shows that fouling and cleaning constraints had

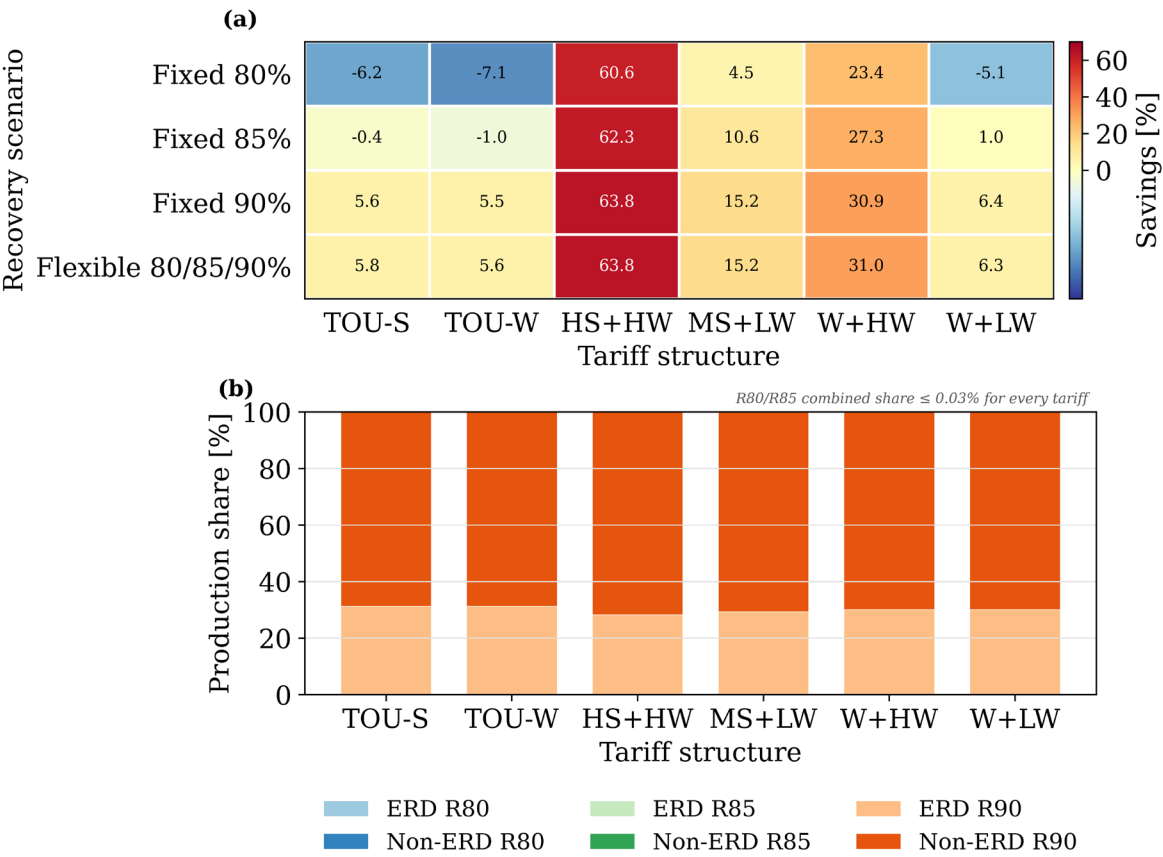
only minor effects on high-volatility RTP cases, although they affected scenarios with limited underlying economic value. Recovery flexibility partially restored the lost value, as demonstrated by the increase in savings from L0 to L5.

#### **Value of Recovery-Ratio Flexibility Under Fixed-Feed Operation**

Recovery-ratio analysis showed that fixed 90% recovery and flexible 80/85/90% recovery outperformed fixed 85% operation under the adopted fixed-feed and fixed-power assumptions. At the constant feed flow of 22.3 ML d<sup>-1</sup> per skid, reducing recovery to 80% decreased permeate production to 17.8 ML d<sup>-1</sup> per skid and therefore required more operating skid-hours to satisfy the 2,650 ML weekly production target. Increasing recovery to 90% raised permeate production to 20.0 ML d<sup>-1</sup> per skid and reduced the required operating skid-hours. Because the calibrated ERD and non-ERD power coefficients were held constant across recovery modes, higher recovery also increased the modeled permeate production per unit of assigned skid energy. Recovery-dependent fouling partially counteracted this production advantage but did not fully offset it under the base-case parameterization (Figure 4a).

The flexible 80/85/90% case produced the highest savings in five of the six tariff structures, with savings of 63.8% under RTP HS+HW, 31.0% under RTP W+HW, 15.2% under RTP MS+LW, 6.3% under RTP W+LW, 5.8% under TOU-Summer, and 5.6% under TOU-Winter. However, the improvement of flexible 80/85/90% operation over fixed 90% recovery was negligible. This indicates that the optimizer selected 90% recovery during nearly all operating periods because the additional permeate produced at fixed feed flow and assigned skid power outweighed the higher modeled fouling multiplier. The result therefore reflects the combined influence of the fixed-power assumption and the selected fouling formulation, rather than demonstrating that 90% recovery is inherently the most energy-efficient or operationally preferable condition at the full-scale facility.

691 In the flexible recovery scenario, the optimizer assigned nearly all production to 90% recovery  
692 (Figure 4b). Across the tariff structures, approximately 28-31% of production came from ERD  
693 skids at 90% recovery, while approximately 69-72% came from non-ERD skids at 90% recovery.  
694 Production at 80% recovery was not selected, and production at 85% recovery was negligible. This  
695 result suggests that, under the present assumptions, recovery flexibility behaves primarily as a  
696 productivity-enhancing option rather than as a dynamic tradeoff between energy, fouling, and  
697 recovery.  
698



**Figure 4.** Effect of recovery-ratio selection on weekly cost savings and optimized production allocation under fixed-feed and fixed-power assumptions. (a) Weekly savings for fixed 80%, fixed 85%, fixed 90%, and flexible 80/85/90% recovery scenarios. (b) Permeate-production shares assigned to ERD and non-ERD skids at each recovery mode in the flexible scenario. Recovery changed permeate production and the fouling multiplier, while configuration-specific per-skid power coefficients remained constant.

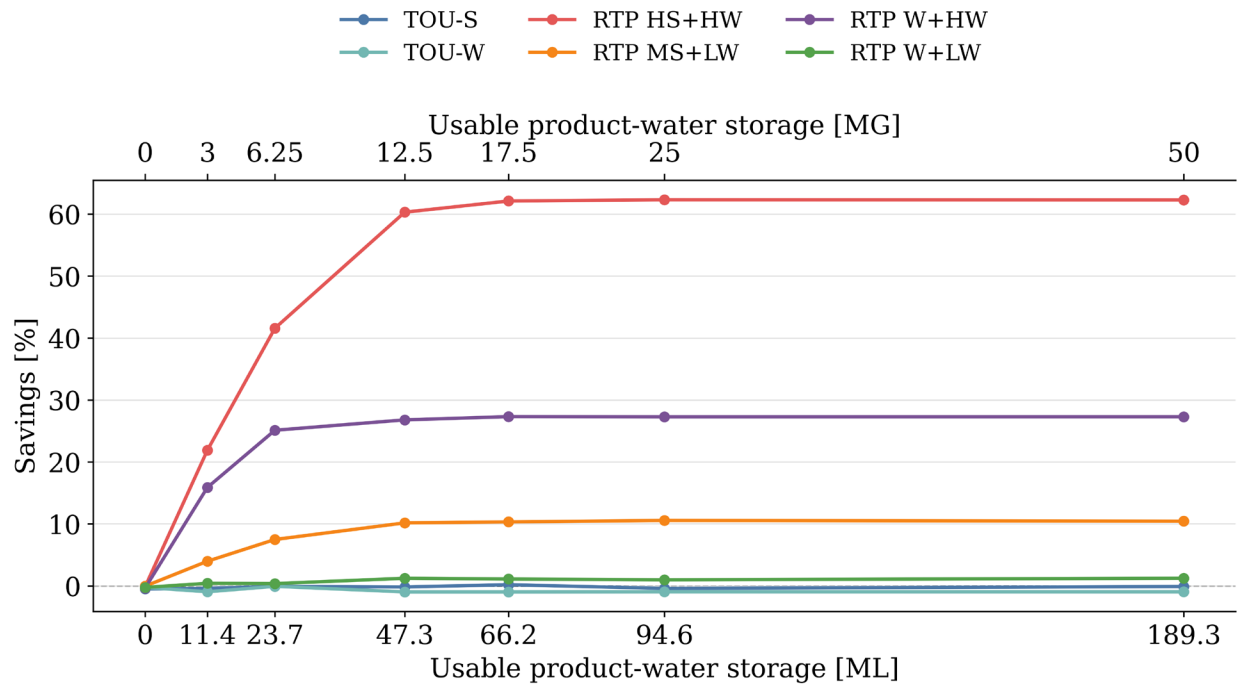
This result should be interpreted within the assumptions of the recovery-flexible formulation. It does not demonstrate that 90% recovery is always preferable in full-scale operation. Because feed

flow and per-skid power were held constant across recovery modes, increasing recovery provided more permeate without an explicit increase in modeled pressure or instantaneous power demand. Although the fouling multiplier increased at higher recovery, the associated cleaning and availability effects were not sufficient to offset this modeled production advantage. The strong preference for 90% recovery therefore reflects both the fixed-power assumption and the current fouling parameterization. Future model development should incorporate site-specific relationships among recovery, feed pressure, interstage-booster demand, ERD performance, permeate quality, scaling potential, permeability decline, pressure drop, and cleaning frequency.

#### **Effect of Product-Water Storage Capacity on Load-Shifting Potential**

Storage capacity was a critical enabler of tariff-responsive operation under volatile RTP structures. With no storage, savings were near zero or slightly negative for all tariffs, ranging from -0.5% to -0.1% (Figure 5). Adding 23.7 ML of storage, equivalent to approximately 1.5 h of average demand at 378.5 ML d<sup>-1</sup>, increased savings to 41.6% under RTP HS+HW and 25.1% under RTP W+HW. Increasing storage to 47.3 ML further improved the RTP HS+HW savings to 60.3%, while 94.6 ML increased savings to 62.3%.

Additional storage beyond 94.6 ML provided little additional benefit for the high-volatility RTP cases. Under RTP HS+HW, savings were 62.3% and 62.3% for 94.6 and 189.3 ML, respectively. This suggests that, for the one-week tariff profiles studied here, approximately 94.6 ML of storage captured most of the available load-shifting value.



**Figure 5.** Influence of product-water storage capacity on tariff-responsive RO savings. Savings are shown as a function of usable downstream storage capacity for each tariff structure.

The diminishing marginal value of increasing storage capacity was expected because the number and duration of economically favorable load-shifting periods within each weekly tariff profile are finite. The more informative result is the relatively modest storage capacity required to capture most of this value. Under RTP HS+HW, increasing storage from zero to 23.7 ML increased savings to 41.6%, while 47.3 ML increased savings to 60.3%, compared with 62.3% at 94.6 ML. Increasing storage further to 189.3 ML produced no additional savings. Thus, although diminishing returns with increasing storage were anticipated, the analysis identifies an approximate storage threshold

beyond which additional capacity provides little incremental economic benefit under the representative weekly price signals considered here.

# **Influence of Demand Charges on Tariff-Responsive Operation**

Demand-charge sensitivity showed that peak-demand penalties substantially reduced economic savings, particularly under RTP tariffs. With no demand charge, savings reached 67.6% under RTP HS+HW, 36.2% under RTP W+HW, and 18.2% under RTP MS+LW. At the base-case demand charge of 20 USD/kW-month, savings declined to 62.3%, 27.3%, and 10.6%, respectively. At 60 USD/kW-month, savings declined further to 54.4%, 19.1%, and 4.2%.

**Table 5.** Demand-charge sensitivity.

| Demand charge   | Savings [%] |       |              |              |             |             |
|-----------------|-------------|-------|--------------|--------------|-------------|-------------|
|                 | TOU-S       | TOU-W | RTP<br>HS+HW | RTP<br>MS+LW | RTP<br>W+HW | RTP<br>W+LW |
| 0 USD/kW-month  | 5.3         | 2.8   | 67.6         | 18.2         | 36.2        | 7.7         |
| 10 USD/kW-month | 1.7         | −0.6  | 64.8         | 13.1         | 31.1        | 3.3         |
| 20 USD/kW-month | −0.4        | −1.0  | 62.3         | 10.6         | 27.3        | 1.0         |
| 40 USD/kW-month | −0.35       | −0.1  | 57.6         | 5.8          | 22.7        | −0.4        |
| 60 USD/kW-month | −0.1        | −0.1  | 54.4         | 4.2          | 19.1        | −0.9        |

These results confirm that energy-only analyses overestimate the practical value of flexible RO operation. Demand charges penalize concentrated production during low-energy-price hours by

increasing the cost of peak power. Nevertheless, under the most volatile RTP structures, large savings persisted even at high demand-charge rates, indicating that flexible operation can remain valuable when price spikes are sufficiently severe.

#### Sensitivity of Optimized Operation to Fouling and Cleaning Assumptions

For the fixed-85% recovery case, fouling-interval and cleaning-duration sensitivities had relatively small impacts on savings. Across fouling intervals of 360, 720, and 1440 h, savings under RTP HS+HW remained 62.3%, and savings under RTP W+HW remained 27.3%. Cleaning duration had similarly limited influence, with RTP HS+HW savings ranging from 62.1% to 62.3% across 2-8 h cleaning durations.

**Table 6.** Fouling-interval and cleaning-duration sensitivity.

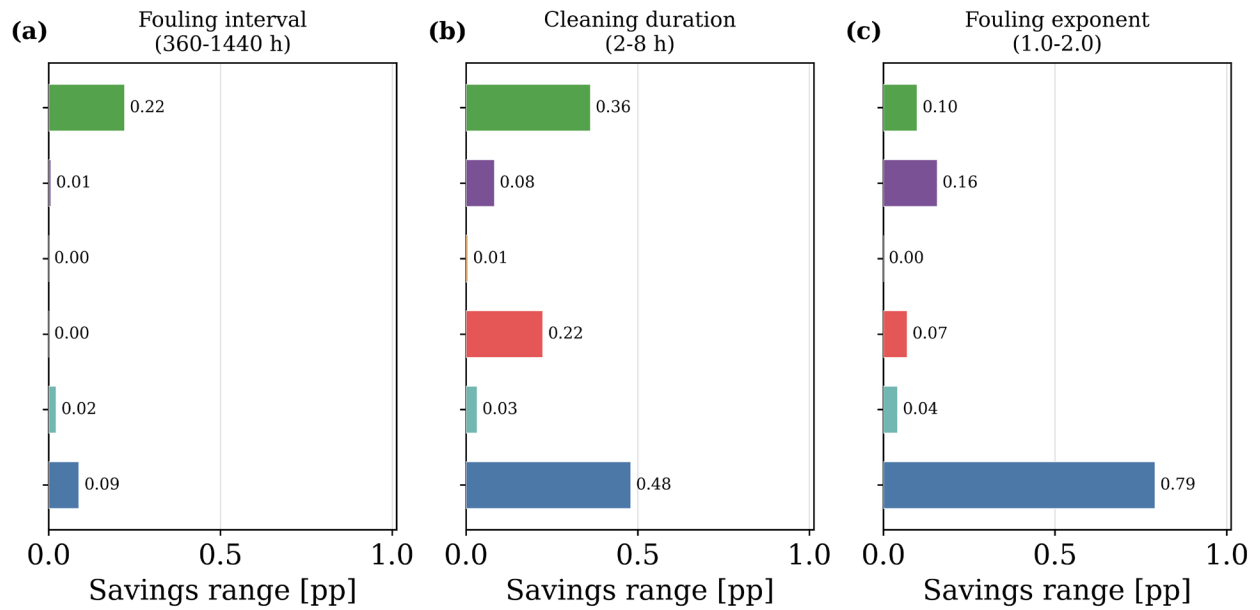
| Sensitivity case          | Savings [%] |       |              |              |             |             |
|---------------------------|-------------|-------|--------------|--------------|-------------|-------------|
|                           | TOU-S       | TOU-W | RTP<br>HS+HW | RTP<br>MS+LW | RTP<br>W+HW | RTP<br>W+LW |
| Fouling interval = 360 h  | -0.4        | -1.0  | 62.3         | 10.6         | 27.3        | 0.8         |
| Fouling interval = 720 h  | -0.4        | -1.0  | 62.3         | 10.6         | 27.3        | 1.0         |
| Fouling interval = 1440 h | -0.4        | -1.0  | 62.3         | 10.6         | 27.3        | 0.8         |
| Cleaning duration = 2 h   | -0.6        | -1.0  | 62.3         | 10.6         | 27.3        | 0.8         |
| Cleaning duration = 4 h   | -0.4        | -1.0  | 62.3         | 10.6         | 27.3        | 1.0         |
| Cleaning duration = 8 h   | -0.1        | -0.9  | 62.1         | 10.6         | 27.2        | 1.1         |

771 This limited sensitivity occurs because, in the fixed-85% case, all online skids accumulate fouling  
772 at the same recovery-dependent multiplier. Thus, fouling primarily affects skid availability and  
773 cleaning scheduling rather than changing the optimizer's recovery selection. However, fouling  
774 becomes more important in recovery-flexible cases because 90% recovery has a larger  
775 concentration factor and therefore a higher fouling multiplier.

776 In the recovery-flexible sensitivity, changing the fouling exponent from 1.0 to 2.0 had only a  
777 modest effect on total savings. The flexible recovery case still achieved 63.8-63.8% savings under  
778 RTP HS+HW and 30.8-31.0% under RTP W+HW. This again indicates that, under the current  
779 fouling-cost assumptions, the benefit of higher recovery remains dominant.

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**Figure 6.** Sensitivity of optimized cost savings to fouling and cleaning model parameters. Each panel isolates one fouling/cleaning parameter and re-solves the MILP at three tested values, holding every other model setting at its reference configuration (storage balance, demand charge, startup/shutdown penalties, and fouling dynamics all active). Bars show the resulting range in optimized savings (maximum – minimum across the three tested values, in percentage points, pp) for each of the six tariff structures, a longer bar indicates the cost-optimal solution is more sensitive to that parameter, with the exact range labeled at each bar tip. (a) Fouling interval, the reference cleaning-cycle length at 85% recovery, swept over 360, 720, and 1440 h with recovery fixed at 85%. (b) Cleaning duration swept over 2, 4, and 8 h, also with recovery fixed at 85%. (c) The fouling-recovery exponent ( $\gamma$ , governing how strongly concentration factor drives fouling rate) swept over 1.0, 1.5, and 2.0.

## **Impact of Startup and Shutdown Costs on Dispatch Flexibility**

The operationally constrained reference case model included the electricity consumed during startup and shutdown transitions but assigned no additional fixed monetary penalty to individual cycling events. To evaluate the possible influence of equipment wear, maintenance, labor, and other unquantified cycling-related impacts, fixed startup and shutdown costs were subsequently introduced through a sensitivity analysis. The tested startup/shutdown penalty pairs were 0/0, 25/5, 50/10, 100/20, and 250/50 USD per skid. The 0/0 USD per-skid case therefore corresponds to the base-model assumption.

Increasing the fixed penalty pair from 0/0 to 250/50 USD per skid reduced savings under RTP HS+HW from 62.3% to 56.0% and under RTP W+HW from 27.3% to 13.7%. The moderate-summer RTP case was more sensitive, with savings decreasing from 10.6% to -1.0%. Conventional TOU cases remained near zero or negative across the tested cycling-cost assumptions (Table 7).

**Table 7.** Sensitivity of weekly cost savings to assumed fixed startup and shutdown penalty costs. The 0/0 USD per-skid case represents the base model, in which startup and shutdown events incur transition-energy costs but no additional fixed monetary penalty.

| Startup/shutdown<br>fixed cost [USD] | Savings [%] |       |              |              |             |             |
|--------------------------------------|-------------|-------|--------------|--------------|-------------|-------------|
|                                      | TOU-S       | TOU-W | RTP<br>HS+HW | RTP<br>MS+LW | RTP<br>W+HW | RTP<br>W+LW |
| 0/0                                  | −0.4        | −1.0  | 62.3         | 10.6         | 27.3        | 1.0         |
| 25/5                                 | −1.0        | −1.0  | 61.5         | 8.5          | 26.2        | −0.6        |
| 50/10                                | −0.1        | −0.2  | 61.0         | 6.1          | 24.8        | −0.8        |
| 100/20                               | −0.1        | −0.1  | 59.5         | 3.6          | 21.6        | −1.0        |
| 250/50                               | −0.4        | −0.4  | 56.0         | −1.0         | 13.7        | −0.3        |

#### Annualized Value under Representative Tariff-Frequency Scenarios

The weekly results showed that flexible RO operation is highly sensitive to tariff structure; however, the most aggressive RTP tariff cases do not represent every week of the year. Therefore, weekly costs were converted into annualized values using equivalent-week tariff-frequency profiles. This analysis provides a more practical estimate of expected annual savings than any individual weekly tariff case alone.

Annualized savings depended strongly on the assumed frequency of high-volatility RTP conditions (Table 8). Under the RTP base profile, the fixed-85% recovery case reduced annual cost from 13.04

824 million USD/yr to 10.40 million USD/yr, corresponding to 2.63 million USD/yr in savings, or  
825 20.2%. Allowing recovery flexibility increased savings to 3.17 million USD/yr, or 24.3%. Under  
826 the RTP conservative profile, annual savings were lower but still substantial, reaching 13.7% for  
827 fixed-85% recovery and 18.2% for flexible 80/85/90% recovery. Under the RTP high-volatility  
828 profile, savings increased to 30.2% and 33.7% for the fixed-85% and recovery-flexible cases,  
829 respectively.

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**Table 8.** Annualized economic value under calendar and RTP tariff-frequency assumptions.

| Annual profile      | Operating scenario          | Annual baseline cost [USD/yr] | Annual optimized cost [USD/yr] | Annual savings [USD/yr] | Annual savings [%] |
|---------------------|-----------------------------|-------------------------------|--------------------------------|-------------------------|--------------------|
| RTP base            | Fixed 85% recovery          | 13,037,700                    | 10,402,839                     | 2,634,861               | 20.2               |
| RTP base            | Flexible 80/85/90% recovery | 13,037,700                    | 9,867,408                      | 3,170,292               | 24.3               |
| RTP conservative    | Fixed 85% recovery          | 11,848,641                    | 10,221,214                     | 1,627,427               | 13.7               |
| RTP conservative    | Flexible 80/85/90% recovery | 11,848,641                    | 9,687,699                      | 2,160,942               | 18.2               |
| RTP high-volatility | Fixed 85% recovery          | 15,415,817                    | 10,766,090                     | 4,649,727               | 30.2               |
| RTP high-volatility | Flexible 80/85/90% recovery | 15,415,817                    | 10,226,824                     | 5,188,993               | 33.7               |
| TOU calendar        | Fixed 85% recovery          | 10,559,434                    | 10,640,903                     | −81,468                 | −0.8               |
| TOU calendar        | Flexible 80/85/90% recovery | 10,559,434                    | 9,962,438                      | 596,996                 | 5.7                |

The TOU calendar profile produced a contrasting result. Under fixed-85% recovery, annual operation was slightly more expensive than the steady-state baseline, with annual savings of −0.8%. This is consistent with the weekly TOU results, where conventional TOU-8 price spreads were not large enough to compensate for operational constraints. In contrast, the recovery-flexible case achieved 5.7% annual savings under the TOU calendar profile, equal to approximately 0.60

million USD/yr. This indicates that recovery flexibility can provide economic value even when temporal price volatility is limited, primarily by increasing permeate production per online skid.

Recovery flexibility improved annual savings in all annualization profiles. The additional annual savings from flexible 80/85/90% recovery relative to fixed-85% operation were approximately 0.54 million USD/yr under the RTP base, conservative, and high-volatility profiles, and approximately 0.68 million USD/yr under the TOU calendar profile. These results show that the practical value of flexible RO operation depends on both the magnitude of weekly savings and the frequency of high-value tariff conditions. Therefore, annualized assessment is essential for translating weekly optimization results into plant-level decision support.

#### **Cost-component Breakdown and Operational Dispatch Patterns**

The optimized cost was dominated by energy and demand-charge components. In the operationally constrained fixed-85% reference case, energy cost represented approximately 76.6-81.6% of optimized total cost, while demand charges represented approximately 18.1-23.1%. Transition-energy costs were small, accounting for less than 1% of the optimized total cost under all tariff structures. Because fixed startup and shutdown penalties were set to zero in the base model, the transition-cost percentages reported in Table 9 represent only the additional electricity consumed during startup and shutdown events. Cleaning costs were zero in the base case because no fixed or energy-based cleaning cost was assigned; however, cleaning still constrained the optimization by reducing available skid capacity during cleaning periods and enforcing terminal fouling limits.

Operationally, the optimized schedules relied on large changes in online skid count under RTP tariffs. For example, under the RTP HS+HW tariff, the number of online skids ranged from 3 to 25, indicating substantial production shifting away from high-price hours. In contrast, under TOU-

862 8 winter, the optimized schedule remained close to the baseline, with 19-21 skids online and  
863 negligible economic benefit. This contrast highlights that the practical value of flexible RO  
864 operation is determined by the interaction between tariff volatility, storage capacity, demand-  
865 charge exposure, and allowable skid cycling.

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## DISCUSSION

### **Tariff-dependent value of full-scale RO flexibility**

The tariff-dependent results support the first hypothesis that flexible RO operation provides limited economic value when temporal electricity-price differences are small but becomes substantially more valuable under sufficiently volatile price signals. Under the operationally constrained fixed-85% reference case, optimized operation produced negligible or slightly negative savings under the conventional TOU-8 Summer and Winter profiles, whereas savings increased to 10.6%, 27.3%, and 62.3% under the RTP MS+LW, W+HW, and HS+HW profiles, respectively. The qualitative dependence of flexibility value on electricity-price variability was expected; the more informative result is the magnitude of the contrast obtained for the same physical facility, ranging from a slight increase in weekly operating cost to a reduction exceeding 60% under the most volatile representative price signal.

These results reinforce the distinction between operational flexibility and economically valuable flexibility. A facility may possess sufficient treatment capacity and storage to shift production, yet flexible dispatch provides little benefit when electricity-price differences are insufficient to offset demand charges, transition energy, and other operational constraints. Conversely, highly variable price profiles can create sufficiently large avoided-cost opportunities to justify substantial production shifting. This interpretation is consistent with previous water-sector and desalination optimization studies showing that tariff structure, electricity-market conditions, and the temporal alignment of water production with electricity prices strongly influence the economic value of flexible operation (Ghorbani Bam et al., 2026; Rao et al., 2024a).

The optimized dispatch under RTP HS+HW illustrates the operational mechanism responsible for the largest savings. Online skid count varied from 3 to 25, allowing production to be substantially reduced during high-price periods and restored during lower-cost hours. By comparison, the TOU-8 Winter schedule remained close to the 20-skid steady-state baseline, with 19–21 skids online. Flexible operation should therefore not necessarily be implemented as a continuously active strategy under every tariff condition. A more practical approach may be selective or event-responsive deployment, in which aggressive production shifting is enabled only when forecasted electricity-price differences exceed the operational cost and risk associated with cycling the facility.

#### **Effects of storage, demand charges, and operational constraints**

The constraint-isolation analysis showed that finite storage alone caused almost no reduction relative to the ideal dispatch case once 94.6 ML of usable storage was available. This finding supports the second hypothesis that storage is an important enabling constraint, but it also provides an important qualification: the economic value of additional storage does not increase proportionally with storage capacity. The qualitative occurrence of diminishing returns was expected because the number and duration of favorable load-shifting periods within each weekly price profile are finite. The more informative result is how rapidly the available flexibility value was captured. Under RTP HS+HW, savings increased from approximately zero without storage to 41.6% with 23.7 ML and 60.3% with 47.3 ML, compared with 62.3% at 94.6 ML; doubling storage further to 189.3 ML produced no additional savings. Thus, storage was essential for enabling load shifting, but relatively moderate capacity captured most of the available economic opportunity under the representative high-volatility price signal.

910 The economic interpretation depends, however, on whether storage capacity is already available.  
911 The model treated 94.6 ML as usable downstream storage and did not include the capital cost of  
912 constructing, expanding, or modifying a storage facility. If this capacity is already available within  
913 the OCWD system and can be operationally allocated to RO load shifting, its original construction  
914 cost is a sunk cost and should not be charged entirely to the flexible-operation strategy. In contrast,  
915 facilities requiring a new dedicated tank, additional pumping, transmission piping, controls, land,  
916 or seismic upgrades would need to compare annual operating savings with the annualized capital  
917 and operating costs of that infrastructure. Generic storage-cost factors would be unreliable because  
918 tank type, site conditions, land availability, foundation requirements, hydraulic integration, and  
919 regional construction costs can materially affect project cost. Engineering cost models therefore  
920 generally rely on component-level and site-specific design inputs rather than a single transferable  
921 cost per unit of storage.

922 The constraint hierarchy broadly supported the second hypothesis, with peak-demand exposure  
923 emerging as the dominant modeled economic restriction on energy arbitrage. Introducing the  
924 demand-charge component between L2 and L3 reduced RTP HS+HW savings from 67.7% to  
925 62.6% and RTP W+HW savings from approximately 36.6% to 27.7%. The dedicated sensitivity  
926 analysis produced the same conclusion: increasing the demand-charge rate from 0 to 60 USD kW<sup>-1</sup>  
927 month<sup>-1</sup> reduced savings from 67.6% to 54.4% under RTP HS+HW and from 36.2% to 19.1%  
928 under RTP W+HW. Concentrating production into low-energy-price hours can therefore reduce  
929 volumetric electricity charges while simultaneously increasing the maximum electrical load used  
930 to determine demand charges. Analyses that consider only hourly energy prices may consequently  
931 overestimate the economic value of flexible RO operation.

This tradeoff suggests that future control strategies should co-optimize energy charges and peak demand rather than respond only to hourly electricity prices. Depending on the billing structure, it may be preferable to preserve a moderate production level during high-price periods if complete curtailment would require a larger rebound peak later in the billing period. Additional flexibility resources, such as coordinated operation of other treatment processes, on-site energy storage, or explicit peak-power constraints, could further reduce this rebound effect.

Startup and shutdown transition energy had a comparatively small effect in the base model, accounting for less than 1% of optimized cost. This result should not be interpreted as evidence that frequent cycling is operationally costless. The base formulation included only the additional electricity consumed during the assumed 200-s startup and 60-s shutdown periods; it did not assign fixed costs for operator supervision, flushing, equipment wear, maintenance, or accelerated component replacement. The fixed-cost sensitivity showed that these omitted effects could be important. At assumed startup/shutdown costs of 250/50 USD per skid, savings declined from 62.3% to 56.0% under RTP HS+HW and from 27.3% to 13.7% under RTP W+HW, while the RTP MS+LW case became uneconomic. High-volatility tariffs therefore remained comparatively robust, whereas moderate-price cases were sensitive to costs that are currently uncertain.

#### **Recovery flexibility and membrane-performance implications**

The recovery-flexibility results only partially supported the third hypothesis. As expected, allowing recovery to vary provided additional modeled flexibility by changing permeate production per operating skid. However, the anticipated dynamic tradeoff among recovery, energy use, and membrane impacts did not emerge: the optimizer selected 90% recovery during nearly all operating periods, and flexible 80/85/90% operation provided negligible additional benefit over fixed 90% recovery. This strong preference for the highest recovery should not be interpreted as

evidence that 90% recovery is optimal for OCWD. Feed flow and configuration-specific per-skid power were held constant across recovery modes, so increasing recovery increased permeate production without an explicit increase in feed pressure, interstage-booster demand, or instantaneous electrical power. The recovery-dependent fouling multiplier only partly offset this structural advantage. The result is therefore informative primarily because it reveals the sensitivity of recovery optimization to the assumed hydraulic and membrane-performance representation.

Higher recovery increases the concentration of retained constituents along the membrane stages and can intensify concentration polarization and inorganic scaling potential. Fouling and scaling reduce membrane permeability, increase required pressure, and can increase chemical-cleaning frequency and shorten membrane service life. Recent full-scale and membrane-autopsy literature identifies fouling and scaling as continuing operational challenges in both seawater and brackish-water RO systems, while high-recovery operation may require specialized staging, flux control, pretreatment, and monitoring to avoid unacceptable membrane deterioration (Ahmed et al., 2023; Miyakawa et al., 2021; Philibert et al., 2024).

The present fouling model represented these effects through an aggregate recovery-dependent inventory and cleaning downtime. We did not assign direct costs to cleaning chemicals, waste disposal, replacement membranes, maintenance labor, or lost production beyond the modeled unavailability period. Consequently, the economic advantage of 90% recovery may need a conservative revision. More frequent cleaning would increase chemical and labor costs and could require additional operator coordination, while cleaning-related downtime may reduce the flexibility available during high-value tariff events. Cleaning itself can also contribute to membrane aging when performed frequently or under aggressive chemical conditions.

Future recovery-flexibility analysis should therefore use recovery-specific hydraulic and membrane-performance coefficients. Ideally, feed pressure, booster-pump power, ERD performance, concentrate flow, salt passage, scaling indices, normalized permeability decline, and cleaning interval would be calibrated from OCWD operating data or a validated membrane-process simulator. A more complete economic formulation would assign costs to cleaning chemicals, labor, waste handling, replacement membranes, and lost treatment availability. Until these relationships are incorporated, the fixed-85% results provide a stronger basis for operational decision-making than the recovery-flexible results.

#### **Annualized value and practical implementation**

The annualization analysis supports the temporal-scale component of the third hypothesis and demonstrates why extreme weekly savings should not be extrapolated directly across an entire year. Under the fixed-85% formulation, annual savings ranged from 13.7% in the conservative RTP profile to 30.2% in the high-volatility profile, with an intermediate value of 20.2%, or approximately USD 2.63 million, under the RTP base profile. In contrast, the conventional TOU calendar produced a slight annual cost increase of 0.8%. Thus, annual economic value depends not only on the magnitude of savings achievable during an individual tariff condition but also on how frequently favorable price conditions occur.

The equivalent-week profiles should therefore be interpreted as **representative potential alternative tariff-frequency scenarios**, rather than as predictions of OCWD's actual electricity contract or future electricity expenditures. OCWD currently purchases electricity through a third-party arrangement, whereas the SCE profiles were used here as regional price signals representing alternative levels of temporal price variability. In addition, the annualization procedure aggregated representative weekly costs without preserving their chronological order. Seasonal transitions,

1000 month-to-month peak-demand interactions, interweek storage and fouling states, and contractual  
1001 provisions specific to the third-party supplier were therefore not reproduced. A future assessment  
1002 based on a full year of chronological prices and actual contractual billing rules would provide a  
1003 stronger estimate of realizable annual savings.

1004 Implementation would also require consideration of operational reliability and uncertainty. The  
1005 deterministic formulation assumes perfect knowledge of hourly electricity prices, constant  
1006 downstream demand of 378.5 ML d<sup>-1</sup>, and successful execution of the optimized skid schedule. In  
1007 practice, downstream water demand varies temporally, and its relationship with the electricity-  
1008 price signal could materially affect both the required storage capacity and the value of flexible  
1009 operation. For example, elevated water demand during high-price periods would require greater  
1010 withdrawal from storage when RO production is economically curtailed and could therefore  
1011 increase the storage capacity needed for load shifting. Conversely, if higher demand coincides with  
1012 low-price periods when RO production is increased, less intermediate storage may be required.  
1013 Actual facilities also experience electricity-price forecast errors, unplanned equipment outages,  
1014 variable feedwater quality, maintenance restrictions, and operator-imposed reliability margins.  
1015 Large hourly changes in skid count may further affect upstream pretreatment, downstream  
1016 UV/AOP operation, hydraulic transients, and product-water quality. Model-predictive or rolling-  
1017 horizon control using updated price, demand, storage, and equipment-availability forecasts would  
1018 therefore be more appropriate for field deployment than a fixed week-ahead schedule. The  
1019 annualized demand-charge component should similarly be interpreted as a comparative  
1020 approximation rather than a reconstruction of chronological monthly utility bills.

1021 The analysis focused on electricity cost and did not evaluate greenhouse-gas emissions. Shifting  
1022 demand to lower-price hours does not necessarily shift it to hours with lower marginal emission

intensity, particularly when prices are influenced by congestion, scarcity, or contractual factors. Future work should therefore examine the joint optimization of operating cost, marginal emissions, and water-system reliability rather than assuming that economic and environmental objectives are always aligned.

### **Economic limitations and recommended investment analysis**

The reported savings are operating-cost savings and should not be interpreted as net project returns. In addition to potential storage investment, implementation may require supervisory control modifications, communications infrastructure, cybersecurity measures, instrumentation, operator training, and modifications to pumps, valves, or process controls. Increased startup frequency, higher recovery, and additional cleaning can also create recurring maintenance and labor costs that were not quantified.

Because storage costs depend strongly on tank type, land availability, hydraulic integration, foundation requirements, pumping infrastructure, and regional construction conditions, a generic unit-capacity estimate was not applied. Future work should evaluate project-specific net present value and break-even capital investment using the annual operating savings estimated here. The annual net benefit could be expressed as:

$$B_{\text{net}} = S_{\text{electricity}} - C_{\text{storage,annualized}} - C_{\text{controls}} - C_{\text{additional O\&M}} - C_{\text{cleaning}} - C_{\text{cycling}}$$

(43)

where the annualized capital component is calculated using the expected project life and discount rate. This framework would allow the analysis to distinguish between the use of existing storage, which may require only control and operational changes, and construction of new dedicated storage. A screening analysis could also report the maximum capital investment that would be

1045 justified by the modeled annual savings under each tariff-frequency profile, without requiring a  
1046 generalized tank-cost estimate that may not represent site-specific design and construction  
1047 conditions.

1048

## CONCLUSIONS

A full-scale mixed-integer optimization framework was developed to evaluate tariff-responsive operation of the OCWD RO system while accounting for heterogeneous ERD and non-ERD skids, finite product-water storage, peak-demand charges, startup and shutdown transitions, fouling, cleaning, and weekly production requirements. The analysis produced five principal conclusions.

First, flexible RO operation was not universally cost-effective. Under the operationally constrained fixed-85% reference case, savings were slightly negative under conventional TOU-8 tariffs but reached 10.6–62.3% under the evaluated RTP profiles. Large savings therefore require sufficiently strong and sustained electricity-price differentials.

Second, product-water storage enabled load shifting, but its value exhibited strong diminishing returns. Approximately 94.6 ML captured nearly all the available flexibility for the representative weeks examined. This result applies directly when such capacity is operationally available; facilities requiring new storage must account for capital, pumping, land, and maintenance costs.

Third, demand charges were the dominant modeled limitation on flexibility. Startup and shutdown transition energy and the tested fouling parameters had smaller effects in the base model, although assumed fixed cycling costs substantially reduced the value of moderate-volatility tariffs. Flexible operation is therefore most robust when deployed selectively during high-value pricing events rather than indiscriminately throughout the year.

Fourth, the recovery-flexible formulation favored 90% recovery, but this result was driven partly by the assumption of constant per-skid power across recovery modes. Higher recovery may increase pressure requirements, scaling, cleaning frequency, chemical consumption, labor, downtime, and membrane replacement costs. The recovery-flexibility results should therefore be

1071 viewed as an exploratory upper-bound assessment until recovery-specific hydraulic and  
1072 maintenance data are incorporated.

1073 Fifth, scenario-based annualized savings under the operationally constrained fixed-85% reference  
1074 case ranged from  $-0.8\%$  under the conventional TOU calendar to  $13.7\text{--}30.2\%$  under the RTP  
1075 profiles. These findings show that the practical value of full-scale flexible RO operation depends  
1076 on the interaction among tariff volatility, event frequency, storage availability, demand-charge  
1077 exposure, and allowable operational cycling.

1078 Future work should validate the optimized schedules using chronological electricity-contract data  
1079 and full-scale operating records, incorporate storage and control-system investment costs, quantify  
1080 recovery-dependent cleaning and labor requirements, and evaluate cost, emissions, water quality,  
1081 and reliability within a unified operational framework.

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1090    **AUTHORS CREDIT STATEMENT**

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